

CCNA Exam D

Reorganized to Follow Jeremy's IT Lab CCNA 200-301 Course

74 items - questions, correct answers, and source explanations

What changed: The original Exam D items are reorganized into Jeremy's course order so you can review Boson questions immediately after the matching Jeremy topic. The wording of the source explanations is preserved as closely as practical; obvious PDF/OCR text-layer artifacts are normalized for readability.

Lab items: Original questions 1-4 are NetSim lab tasks. The source provides the task prompt but no separate correct-answer/explanation section, so they are reproduced as lab tasks only.

Cross-topic items: A few integrated questions touch multiple subjects. Each appears once under the closest primary Jeremy topic to avoid duplicates.

Selectable/searchable: All rewritten question, answer, and explanation text in this document is real text, not page screenshots.

Course-Order Index

Only Jeremy sections that contain at least one Exam D item are listed. Original Exam D question numbers are retained for cross-reference to the source PDF.

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Day 1 - Network Devices

Included Exam D items: Q60

Original Exam D Question 60

Approximately how many users would a SOHO network typically have? (Select the best answer.)

- A. from 50 to 250
- B. fewer than 10
- C. more than 1,000
- D. from 500 to 1,000

Correct Answer: B

Explanation

A Small Office/Home Office (SOHO) network would typically have fewer than 10 users. SOHO networks are simple because they are generally designed for small businesses and telecommuters. A

Exam C - organized in Jeremy's IT Lab course order

SOHO router can be used to provide network connectivity, which allows the SOHO users to share a single broadband connection. Users can typically connect to the SOHO router over a wired or wireless connection.

A Small and Medium Business (SMB) network would typically have from 50 to 250 users. SMB networks are more complex than SOHO networks and can provide more functionality. Branch offices can often use network devices that are designed for SMB networks.

An enterprise network would typically have hundreds or thousands of users. Enterprise networks are typically very complex and are designed for large organizations. Enterprise networks sometimes use a two-tier or three-tier network architecture. In a three-tier architecture, the core layer, the distribution layer, and the access layer are used. The core layer is used as the network backbone, the distribution layer is used for routing and security, and the access layer is used for authentication and endpoint connectivity. In a two-tier network architecture, which is used in less-complex network environments, the core and distribution layers are combined into a collapsed core layer.

Day 2 - Interfaces & Cables

Included Exam D items: Q5

Original Exam D Question 5

Drag each term to the appropriate transmission medium. All terms should be used.

Terms Single-Mode Fiber Multimode Fiber

Correct Answer: Single-Mode Fiber -> can use a 9-micron core; can use a wavelength of 1,550 nm; typically used for distances greater than 80 km

Multimode Fiber -> can use a 62.5-micron core; can use a wavelength of 850 nm; typically used for distances less than 2 km

Explanation

Multimode fiber (MMF) can use a 62.5-micron core and a wavelength of 850 nanometers (nm). In addition, MMF is typically used for distances less than 2 kilometers (km). When light is transmitted through a fiber-optic cable, the light is only propagated by the fiber core at certain angles, or modes. The light transmitted into the core of an MMF cable is typically in the 850-nm or 1,300-nm frequency range. Because MMF has a relatively large core, 50 micron or 62.5 micron, and the large core permits many different angles of light, the signal becomes dispersed over great distances. This dispersion effectively limits the usable distance of MMF to 2 km. MMF is typically used in campus designs that require at least 1 gigabit per second (Gbps) of bandwidth and network runs less than 2 km.

Single-mode fiber (SMF) typically uses a 9-micron core. In addition, SMF can support a wavelength of 1,550 nm and distances of 80 km. The light transmitted into the core of an SMF cable is typically in the 1,310-nm or 1,550-nm frequency range. Because SMF has a relatively small core that permits very few angles of light, the signal does not become very dispersed over great distances. The limited dispersion of SMF enables network runs of 80 km or more. SMF is typically used in campus designs that require at least 10 Gbps of bandwidth and network runs greater than 2 km.

Because fiber-optic cables transmit light rather than electricity, they are not susceptible to electromagnetic interference (EMI). Therefore, implementing fiber-optic cable can be useful in buildings that contain sources of electrical or magnetic interference.

Days 5-6 - Ethernet LAN Switching

Included Exam D items: Q26, Q52

Original Exam D Question 26

Which of the following fields appears last in an Ethernet frame? (Select the best answer.)

- A. Fcs
- ☐ B. SOF
- C. type
- D. data
- E. preamble
- F. source address
- ☐ G. destination address

Correct Answer: A

Explanation

The 4-byte Frame Check Sequence (FCS) field appears last in an Ethernet frame. The FCS field is a 4-byte cyclic redundancy check (CRC) that is intended to enable a frame's receiver to determine whether the frame has been corrupted in transit. For example, a faulty cable could result in signal degradation and frame corruption. The FCS is calculated based on the values of every other field in the frame. If a CRC error is detected, the frame will be discarded and the interface CRC and Frame counters will be incremented. Similarly, the frame will be discarded if it contains fewer than 64 bytes; a frame containing fewer than 64 bytes is referred to as a runt.

An Ethernet frame typically consists of seven fields in the following order:

- © A7-byte preamble field
- © A 1-byte start-of-frame (SOF) field
- © A6-byte destination address field
- « A6-byte source address field
- © A2-byte type field
- A data field in the range from 46 through 1,500 bytes
- ° A4-byte FCS field

The first five fields of the frame are known as the Ethernet header. The preamble field is used to notify receiving hosts that a frame is being sent. The SOF field is used for synchronization with other hosts on the local area network (LAN). The destination address field contains the Media Access Control (MAC) address of the host for which the data is intended. The source address field contains the MAC address of the host that is sending the data. Finally, an Ethernet header contains a 2-byte type field to indicate the protocol that is intended to receive the frame's data after processing. A payload field of a size in the range from 46 through 1,500 bytes immediately follows the Ethernet header.

Original Exam D Question 52

Which of the following addresses will a switch use to make forwarding decisions? (Select the best answer.)

- A. source MAC addresses
- B. destination MAC addresses
- C. destination IP addresses
- D. source IP addresses

Correct Answer: B

Explanation

A switch will use destination Media Access Control (MAC) addresses to make forwarding decisions. Switches make forwarding decisions based on the destination MAC address contained in a frame's header. The switch first searches the Content Addressable Memory (CAM) table for an entry that matches the frame's destination MAC address. The CAM table, which is also called the switching table, is used by a switch to discover the relationship between the Open Systems Interconnection (OSI) Layer 2 address of a device and the physical port used to reach the device. If the frame's destination MAC address is not found in the table, the switch forwards the frame to all its ports, except the port from which it received the frame. If the destination MAC address is found in the table, the switch forwards the frame to the appropriate port. The source MAC address is also recorded if it did not previously exist in the CAM table.

A switch will not use source MAC addresses to make forwarding decisions. However, a switch will use source MAC addresses to populate the CAM table by recording the source MAC address of an inbound Layer 2 frame and the corresponding switch port that the frame arrived on.

A switch will not use source or destination Internet Protocol (IP) addresses to make forwarding decisions. IP addresses are OSI Layer 3 addresses and are typically used by devices such as routers to make forwarding decisions.

Days 7-8 - IPv4 Addressing

Included Exam D items: Q56

Original Exam D Question 56

Which of the following networks is a globally routable IP address? (Select the best answer.)

- A. 10.1.1.0
- B. 10.16.10
- C. 172.20.1.0
- D. 192.168.1.0
- E. 172.172.1.0
- F. 192.168.111.0

Correct Answer: E

Explanation

The Internet Protocol (IP) address 172.172.1.0 is a globally routable IP address. Globally routable IP addresses are called public IP addresses. Public IP addresses are maintained by the Internet Assigned Numbers Authority (IANA). In order to use a public, routable, globally unique IP address, you must obtain an available range of IP addresses from IANA or from a third party that has obtained valid IP addresses from IANA.

Public IP addresses are limited because each address must be globally unique. You can limit the number of public IP addresses required for your network by using private IP addresses that have been defined in Request for Comments (RFC) 1918 for internal use. This enables you to configure private IP addresses for your network's internal host computers. You can then implement Network Address Translation (NAT) to translate these private addresses to a minimal number of public addresses for transmission across public networks such as the Internet.

To conserve globally unique IP addresses, three network address ranges are defined as private address ranges by RFC 1918. These network ranges are intended for private local area network (LAN) use and are not routed across the Internet. RFC 1918 defines these private address ranges and provides guidelines for their use. The following list indicates the IP address ranges that are reserved for private, internal use:

© 10.0.0.0 to 10.255.255.255

© 172.16.0.0 to 172.31.255.255
© 192.168.0.0 to 192.168.255.255

Days 11-12 - Routing Fundamentals, Static Routing & Life of a Packet

Included Exam D items: Q3, Q13, Q25, Q48

Original Exam D Question 3

Your company's network includes three Cisco routers: RouterA, RouterB, and RouterC. IP addresses have been configured on all devices in the topology.

You have been tasked with configuring static routes on RouterC. No configurations must be made on RouterA or RouterB. Perform the following tasks:

- © Create a static route that uses RouterA as the gateway of last resort.
- © Specify the destination as the next-hop IP address.
- © Use the default AD.
- © Create a floating static route that uses RouterB as a backup default route.
- © Specify the destination as an interface.
- © Use an AD that is 5 greater than the default route AD.
- © Create a static route that routes all packets destined for the 192.168.0.0/24 network to RouterB.
- © Specify the destination as the next-hop IP address.
- © Use the default AD.

Lab task - no separate correct-answer section in the source PDF.

Original Exam D Question 13

You issue the show ip route command on RouterA and receive the following partial output:

```
RouterA#show ip route  
<output omitted>
```

Gateway of last resort is 10.19.54.28 to network 10.140.0.0

```
O E2 172.150.0.0 [110/5] via 10.19.54.6, 0:01:00, FastEthernet1/2
```

What does the 5 within the brackets in the output indicate? (Select the best answer.)

- A. the routing protocol
- B. the AD
- C. the hop count
- D. the metric

Correct Answer: D

Explanation

The 5 within the brackets in the output indicates the metric; the metric is the second number inside

the brackets. Open Shortest Path First (OSPF) uses cost as a metric and calculates cost based on the bandwidth of an interface: the higher the bandwidth, the lower the cost. When two OSPF paths exist to the same destination, the router will choose the OSPF path with the lowest cost.

The 11c within the brackets in the output indicates the administrative distance (AD); the AD is the first number inside the brackets. When multiple routes to a network exist, a router prefers the routing protocol with the lowest AD. The following list contains the most commonly used default ADs:

Route Source AD
Directly connected route 0

Static route 1

EIGRP summary route 3

eBGP 20
Internal EIGRP 90
IGRP 100
'OSPF 110
ISIS 115
RIP 120
External EIGRP 170
iBGP 200
'Unknown 255

A directly connected route, which has an AD of 0, is preferred over any other route to the same network. If a link to a directly connected network goes down, the route with the next lowest AD will be used. Static routes, which have an AD of 1 by default, are preferred after directly connected routes.

The 0 in the output indicates the routing protocol, which, in this scenario, is OSPF. An R in the output would indicate a Routing Information Protocol (RIP) route, and a D in the output would indicate an Enhanced Interior Gateway Routing Protocol (EIGRP) route.

The 5 in the output does not indicate hop count. OSPF uses cost, not hop count, as a metric. By contrast, RIP uses hop count as a metric. If the first character in the output were an R, then the routing protocol would be RIP and therefore, the hop count is the metric.

Original Exam D Question 25

You issue the following commands on RouterA:

```
RouterA#configure terminal
```

```
RouterA(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.4  
RouterA(config)#ip route 10.0.0.0 255.255.0.0 192.168.1.3  
RouterA(config)#ip route 10.0.0.0 255.255.255.0 192.168.1.2  
RouterA(config)#ip route 10.0.0.0 255.255.255.224 192.168.1.1
```

RouterA receives a packet destined for 10.0.0.24.

To which next-hop IP address will RouterA forward the packet? (Select the best answer.)

- A. 10.0.0.4
- B. 192.168.1.2
- C. 192.168.1.3
- D. 192.168.1.4
- E. 192.168.1.1

Correct Answer: E

Explanation

RouterA will forward the packet to the next-hop Internet Protocol (IP) address of 192.168.1.1. When a packet is sent to a router, the router checks the routing table to see if the next-hop address for the destination network is known. The routing table can be filled dynamically by a routing protocol, or you can configure the routing table manually by issuing the ip route command to add static routes. Static routes are not removed from a routing table when the path becomes unavailable. A dynamic

routing protocol is therefore preferable where possible.

The ip route command uses the syntax ip route prefix mask {ip-address | interface}, where prefix is the network address of the destination network, mask is the subnet mask of the destination network, ip-address is the IP address of the next-hop router, and interface is the local interface to which packets should be sent.

A default route is used to send packets that are destined for a location that is not listed elsewhere in the routing table. For example, the ip route 0.0.0.0 0.0.0.0 192.168.1.4 command specifies that packets destined for addresses not otherwise specified in the routing table are sent to the default next-hop address of 192.168.1.4. A prefix and mask combination of 0.0.0.0 0.0.0.0 specifies any packet destined for any network.

If multiple static routes to a destination are known, the most specific route is used. Therefore, the following rules apply on RouterA:

© Packets sent to the 10.0.0.0 255.255.255.224 network are forwarded to the next-hop address of 192.168.1.1. This includes destination addresses from 10.0.0.0 through 10.0.0.31.

c Packets sent to the 10.0.0.0 255.255.255.0 network, except those sent to the 10.0.0.0 255.255.255.224 network, are forwarded to the next-hop address of 192.168.1.2. This includes destination addresses from 10.0.0.32 through 10.0.0.255.

© Packets sent to the 10.0.0.0 255.255.0.0 network, except those sent to the 10.0.0.0 255.255.255.0 network, are forwarded to the next-hop address of 192.168.1.3. This includes destination addresses from 10.0.1.0 through 10.0.255.255.

© Packets sent to any destination not listed in the routing table are forwarded to the default static route next-hop address of 192.168.1.4.

Because the most specific route to 10.0.0.24 is the route toward the 10.0.0.0 255.255.255.224 network, RouterA will forward a packet destined for 10.0.0.24 to the next-hop address of 192.168.1.1.

Original Exam D Question 48

You issue the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command.

What does the number 10 at the end of the command indicate? (Select the best answer.)

- A. the hop count
- B. the administrative distance
- C. the number of packets allowed
- D. the number of IP addresses allowed
- E. the 192.168.1.10 IP address
- F. the next-hop IP address

Correct Answer: B

Explanation

The number 10 at the end of the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command indicates the administrative distance (AD) for the route. The syntax of the ip route command is ip route prefix mask {ip-address | interface} [distance], where prefix is the network address of the destination network, mask is the subnet mask of the destination network, ip-address is the Internet Protocol (IP) address of the next-hop router, interface is the local interface to which the packets should be sent, and distance is the AD of the route.

When multiple routes to a network exist and a more specific route is not available, the router will choose the route with the lowest AD. Therefore, a static route created by issuing the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command would be preferred over a static route created by

the ip route 10.2.2.0 255.255.255.0 192.168.2.1 20 command. An administrator will sometimes create static routes with different ADs so that a backup static route is available in case the primary static route goes down. A backup static route is often referred to as a floating static route.

If the AD is not specified in the ip route command, a static route uses the default AD of 1. Therefore, a static route created by issuing the ip route 10.2.2.0 255.255.255.0 192.168.0.1 command would be preferred over a static route created by the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command.

The number 10 in the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command does not indicate the number of IP addresses or packets allowed by the route. The static route in this scenario will allow any number of packets from the 10.2.2.0 255.255.255.0 network.

The number 10 in the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command does not indicate the hop count for the route. Hop count is used by Routing Information Protocol (RIP) to indicate the number of routers a packet will traverse from source to destination. However, static routes do not use hop count as a metric.

The number 10 in the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command does not indicate the 192.168.1.10 IP address. Packets destined for the 10.2.2.0 255.255.255.0 network will be sent to the next-hop address of 192.168.1.1.

The number 10 in the ip route 10.2.2.0 255.255.255.0 192.168.1.1 10 command does not indicate the next-hop address. The next-hop address of 192.168.1.1 is used for packets destined for the 10.2.2.0 255.255.255.0 network.

Days 13-15 - Subnetting

Included Exam D items: Q18, Q45, Q49

Original Exam D Question 18

Which of the following binary subnet masks is equivalent to the dotted decimal subnet mask 255.255.255.252? (Select the best answer.)

- A. 14149191.11111111.11111111.11111100
- B. 19991911.11111111.11111111.11110000
- C. 19999999.19911111.11111111.11111000
- D. 19141111.11111111.11111111.11100000

Correct Answer: A

Explanation

The binary subnet mask 11111111.11111111.11111111.11111100 is equivalent to the dotted decimal subnet mask 255.255.255.252. There are three ways to express a subnet mask: binary notation, dotted decimal notation, and Classless Inter-Domain Routing (CIDR) notation. Binary notation uses 1s and 0s. Dotted decimal notation uses decimal numbers from 0 through 255. CIDR notation is a shortcut notation that indicates how many bits in the decimal subnet mask are set to a value of 1.

When converting binary notation to dotted decimal notation, you must convert each eight-digit set, or octet, individually. To convert a binary number to decimal, you should calculate the value of each of the binary digits and add them together. The first digit in an eight-digit binary number is worth 128 in decimal, the second digit is worth 64 in decimal, and so on. For example, the binary number 11111111 is calculated by the following method:

128 (1
+64 (1
\$32 (1

+16 (1
+8 (1
+4(1
+2 (1
+1(1

255

X

X

X

X

X

X

X

X

128)

64)

32)

16)

8)

4)

2)

1)

Knowing the decimal value of each of the binary digits is the key to converting from binary to decimal. If you see a 1 for a binary place value, you should add the corresponding decimal equivalent for that place value. If you see a 0 for a binary place value, you should not add anything.

To convert the binary number 11111100, use the following calculation:

128 (1 x 128)

+64 (1 x 64)

+32 (1 x 32)

+16 (1 x 16)

+8 (1 x 8)

+4 (1x 4)

+ (0 x 2)

+0 (0x 1)

252

Therefore, the binary number 11111111.11111111.11111111.11111100 is equal to the dotted decimal number 255.255.252.0.

A subnet mask in CIDR notation begins with a forward slash and a number that indicates how many digits of a subnet mask are used to specify the network portion of an IP address. When calculating the CIDR notation for a subnet mask, you must determine how many binary digits are set to a value of 1. For example, the binary number 11111111.11111111.11111111.11111100 contains 30 consecutive binary 1s; therefore, the subnet mask in CIDR notation is /30.

The binary subnet mask 11111111.11111111.11111111.11100000 is equivalent to the dotted decimal subnet mask 255.255.255.224. You would use the following calculation to convert the fourth octet 11100000 to decimal:

128 (1 x 128)
+64 (1 x 64)
#32 (1 x 32)

224

The binary subnet mask 11111111.11111111.11111111.11110000 is equivalent to the dotted decimal subnet mask 255.255.255.240. You would use the following calculation to convert the fourth octet 11110000 to decimal:

128 (1 x 128)
+64 (1 x 64)
#32 (1 x 32)
+16 (1 x 16)

248

The binary subnet mask 11111111.11111111.11111111.11111000 is equivalent to the dotted decimal subnet mask 255.255.255.248. You would use the following calculation to convert the fourth octet 11111000 to decimal:

128 (1 x 128)
+64 (1 x 64)
#32 (1 x 32)
#16 (1 x 16)
+8 (1 x 8)

248

Original Exam D Question 45

You are implementing a new network with three departments: Administration, Marketing, and Research. Each department's computers will reside on separate subnets within the 192.168.1.0/24 network.

The company does not intend to hire any new employees in the near future, so your supervisor has instructed you to conserve IPv4 address space. The Marketing department requires 25 host addresses, the Administration department requires 55 host addresses, and the Research department requires 110 host addresses.

Which of the following statements correctly indicate the subnet masks that should be used by each department? (Select the best answer)

- A. Marketing should use /26; Administration and Research should use /25.
- B. Marketing should use /28, Administration should use /27, and Research should use /25.
- C. Marketing and Administration should use /27, and Research should use /26.
- D. Marketing should use /27; Administration and Research should use /26.

Marketing should use /27, Administration should use /26, and Research should use 125.

- F. Marketing and Administration should use /26, and Research should use /25.

Correct Answer: E

Explanation

Marketing should use /27, Administration should use /26, and Research should use /25. You have been instructed to conserve Internet Protocol version 4 (IPv4) address space, so you must create subnets containing a minimal number of IP addresses and still accommodate all necessary hosts. A subnet mask specifies how many bits belong to the network portion of a 32-bit Internet Protocol (IP) address. The remaining bits in the IP address belong to the host portion of the IP address. To determine how many host addresses are defined by a subnet mask, use the formula $2^n - 2$, where n is the number of bits in the host portion of the address. Although it is important to learn the formula for calculating valid host addresses, the following list demonstrates the relationship between subnet masks and valid host addresses:

/16 subnet mask contains $2^{16} - 2 = 65,534$ host addresses
/17 subnet mask contains $2^{15} - 2 = 32,766$ host addresses
« /18 subnet mask contains $2^{14} - 2 = 16,382$ host addresses
/19 subnet mask contains $2^{13} - 2 = 8,190$ host addresses
/20 subnet mask contains $2^{12} - 2 = 4,094$ host addresses
e /21 subnet mask contains $2^{11} - 2 = 2,046$ host addresses
/22 subnet mask contains $2^{10} - 2 = 1,022$ host addresses
/23 subnet mask contains $2^9 - 2 = 510$ host addresses

© /24 subnet mask contains $2^8 - 2 = 254$ host addresses
/25 subnet mask contains $2^7 - 2 = 126$ host addresses
/26 subnet mask contains $2^6 - 2 = 62$ host addresses

© /27 subnet mask contains $2^5 - 2 = 30$ host addresses
/28 subnet mask contains $2^4 - 2 = 14$ host addresses

© /29 subnet mask contains $2^3 - 2 = 6$ host addresses
/30 subnet mask contains $2^2 - 2 = 2$ host addresses

The Marketing department requires 25 host addresses. Therefore, a /27 subnet mask is large enough to accommodate the Marketing department.

The Administration department requires 55 host addresses. Therefore, a /26 subnet mask is large enough to accommodate the Administration department.

The Research department requires 110 host addresses. Therefore, a /25 subnet mask is large enough to accommodate the Research department.

Original Exam D Question 49

You have been asked to summarize your network addresses into a single summarized subnet. The addresses on your network consist of the following addresses:

© 10.1.1.128/27
© 10.1.1.160/27
© 10.1.1.192/27
© 10.1.1.224/27

In order to preserve address space, you should summarize these networks by using the most efficient subnet mask.

Which of the following subnet masks should you use? (Select the best answer.)

A. /26
B. /25

Oc 723
D. /24

Correct Answer: B

Explanation

You should use a /25 subnet mask to summarize the four /27 subnets in this scenario. Summarization involves combining subnets to create a larger subnet. The following list demonstrates the relationship between subnets and Internal Protocol (IP) addresses:

e /20 contains $2^{12} = 4,096$ addresses
© /21 contains $2^{11} = 2,048$ addresses
/22 contains $2^{10} = 1,024$ addresses
/23 contains $2^9 = 512$ addresses
/24 contains $2^8 = 256$ addresses

c /25 contains $2^7 = 128$ addresses

© /26 contains $2^6 = 64$ addresses
/27 contains $2^5 = 32$ addresses

© /28 contains $2^4 = 16$ addresses
/29 contains $2^3 = 8$ addresses
/30 contains $2^2 = 4$ addresses

You can use a "rule of two" to determine how many subnets can be summarized within a larger subnet. Each step up to a larger subnet multiplies the number of available subnets by two. The following list demonstrates how many /27 subnets can be summarized within each larger subnet:

© Two /27 subnets can be summarized within a /26 subnet

c Four /27 subnets can be summarized within a /25 subnet

c Eight /27 subnets can be summarized within a /24 subnet

e Sixteen /27 subnets can be summarized within a /23 subnet...
...and so on.

Because there are four /27 subnets on your network, you must use a /25 or larger subnet to summarize these /27 subnets. Therefore, a /26 subnet cannot summarize the networks in this scenario.

Each of the /27 subnets in this scenario contains 32 addresses. The following address ranges exist on each of the subnets in this scenario:

© 10.1.1.128/27 = 10.1.1.128 through 10.1.1.159
© 10.1.1.160/27 = 10.1.1.160 through 10.1.1.191
© 10.1.1.192/27 = 10.1.1.192 through 10.1.1.223
© 10.1.1.224/27 = 10.1.1.224 through 10.1.1.255

A /25 subnet contains 128 addresses. For example, the 10.1.1.128/25 subnet contains addresses from 10.1.1.128 through 10.1.1.255. This subnet encompasses all of the /27 subnet ranges in this scenario. Therefore, you can use a /25 subnet to summarize these addresses.

A /24 subnet contains 256 addresses. For example, the 10.1.1.0/24 network contains addresses from 10.1.1.0 through 10.1.1.255. This address range is enough to contain two /25 subnets, four /26 subnets, eight /27 subnets, and so on. Although the 10.1.1.0/24 subnet can summarize the /27 subnets in this scenario, a /24 subnet is not as efficient as a /25 subnet.

A /23 subnet contains 512 addresses. For example, the 10.1.1.0/23 network contains addresses from 10.1.1.0 through 10.1.2.255. This address range is enough to contain two /24 subnets, four /25

subnets, eight /26 subnets, and so on. Although the 10.1.1.0/23 subnet can summarize the /27 subnets in this scenario, a /23 subnet is not as efficient as a /25 subnet.

Days 16-18 - VLANs

Included Exam D items: Q2, Q15, Q33, Q47, Q68

Original Exam D Question 2

Sieh | Switch

a - ia

Tasks

Your company's network includes two Cisco switches, SwitchA and SwitchB. The switches are connected over their FastEthernet 0/0 interfaces.

Configure SwitchA and SwitchB with the following parameters:
Explicitly configure a trunk between the two switches. Use a standards-based encapsulation protocol.

- * Configure the trunk to send untagged traffic over VLAN 99.
- * Allow traffic only from the three explicitly configured VLANs and from the native VLAN to be sent over the trunk.

Lab task - no separate correct-answer section in the source PDF.

Original Exam D Question 15

You are implementing common Layer 2 security measures on a Cisco switch. You create a new VLAN with an ID of 4. No devices operate on VLAN 4. Next, you issue the following commands on a switch interface:

```
switchport access vlan 4
switchport nonegotiate
```

Which of the following Layer 2 security measures are you implementing? (Select 2 choices.)

- OC A. enabling port security on an access port
- B. moving the port to an unused VLAN
- C. disabling an unused port
- D. disabling DTP on a port
- E. configuring the port mode manually

Correct Answer: B, D

Explanation

You are disabling Dynamic Trunking Protocol (DTP) on a port when you issue the switchport nonegotiate command while you are implementing common Layer 2 security measures on a Cisco switch. In addition, you are moving the port to an unused virtual local area network (VLAN) by issuing the switchport access vlan 4 command. By default, every network interface on a Cisco switch is an active port. Before you deploy a switch on a network, you should take steps to ensure that every trunk port and access port on the switch is secured and that every unused port on the switch is disabled.

By default, all interfaces on a Cisco switch will use DTP to automatically negotiate whether an interface should be a trunk port or an access port. The transmission of DTP packets over an interface can be exploited by a malicious user to obtain information about the network or to convert an interface that should be an access port into a trunked port. You should issue the switchport nonegotiate command on a manually configured port to prevent any attempts by the switch to

negotiate by using DTP.

Moving an unused port to an unused VLAN creates a logical barrier that prevents rogue devices from communicating on the network should such a device connect to the port. To move an access port to an unused VLAN, you should issue the `switchport access vlan vian-id` command on the port, where `vian-id` is the ID of the unused VLAN. When you move an unused port to an unused VLAN, you should also manually configure the port as an access port by issuing the `switchport mode access` command and shut down the port by issuing the `shutdown` command.

You are not configuring the port mode manually by issuing the commands in this scenario. To manually configure a trunk port, you should first issue the `switchport trunk encapsulation protocol` command in interface configuration mode, where `protocol` is the trunk encapsulation protocol you want to use, and then issue the `switchport mode trunk` command in interface configuration mode. To manually configure an access port, you should issue the `switchport mode access` command in interface configuration mode. Manually configuring interfaces to use either trunk mode or access mode effectively disables DTP and ensures that the traffic on those ports is restricted to the intended purpose. Even so, you should issue the `switchport nonegotiate` command on a manually configured trunk port to prevent any attempts by the switch to negotiate by using DTP, because a manually configured trunk port will continue to send DTP frames.

You are not disabling an unused port by issuing the commands in this scenario. Disabling an unused port creates a barrier that prevents rogue devices from communicating on the network should such a device connect to the port. To disable an unused port on a switch, you should issue the `shutdown` command on that port. To verify that a port is in the shutdown state, you should issue the `show interfaces type number` command, where `type` and `number` specify the interface you want to show. A port that has been shut down will be reported as administratively down by the `show interfaces type number` command.

You are not enabling port security on an access port by issuing the commands in this scenario. To protect switch interfaces against Media Access Control (MAC) flooding attacks, you should enable port security on all access mode interfaces on the switch. Issuing the `switchport port-security` command in interface configuration mode enables port security with default settings. You can modify port security settings before you enable port security by issuing the `switchport port-security mac-address mac-address` command, the `switchport port-security maximum maximum-number-of-mac-addresses` command, and the `switchport port-security violation [protect | restrict | shutdown]` command.

When enabled with its default settings, port security will shut down a port on which a violation occurs. In addition, port security will allow only the first MAC address to connect to the port to access the port.

Original Exam D Question 33

SwitchB

VLAN

172.16.11.1/24 10.1.51.10124

VLAN21

192,168.21.1/24 172.16.31.2130

aa

Siitona 21641124" Internet)

\)

Host RouterA oe eee

172.16.11.13724

HostB
192.168.21.13/24

You administer the network shown above.

Which of the following should you configure as the default gateway on Host to enable HostA to access the Internet? (Select the best answer.)

- A. 172.16.31.1/30
- B. 192.168.21.1/24
- C. 10.1.51.10/24
- D. 10.1.51.5/24
- E. 172.16.41.1/24

OF 172.16.11.1/24

Correct Answer: F

Explanation

You should configure 172.16.11.1/24 as the default gateway on HostA to enable HostA to access the Internet. Hosts in a particular virtual local area network (VLAN) should use the switched virtual interface (SVI) associated with that VLAN as their default gateway. The SVI is a virtual interface that represents all the physical interfaces in the VLAN. In this scenario, VLAN11 is assigned the network range of 172.16.11.0/24. The SVI for VLAN11 has been assigned the Internet Protocol version 4 (IPv4) address 172.16.11.1/24 on SwitchB. Because HostA resides in VLAN11, HostA should use 172.16.11.1/24 as its default gateway.

You should not configure 192.168.21.1/24 as the default gateway for HostA. VLAN21 is assigned the 192.168.21.0/24 network and uses the SVI address 192.168.21.1/24 for its default gateway. HostA is not a member of VLAN21 and cannot access any of the devices in VLAN21 without the assistance of a router. The default gateway must be an address in the same Internet Protocol (IP) network range as HostA.

You should not configure 10.1.51.5/24 or 10.1.51.10/24 as the default gateway for HostA. The 10.1.51.5/24 and 10.1.51.10/24 addresses are the management IP addresses of SwitchA and SwitchB, respectively. HostA is unable to reach the 10.1.51.0/24 network without a static route or a properly assigned default gateway because HostA does not have an IP address in the 10.1.51.0/24 network.

You should not configure 172.16.31.1/30 or 172.16.41.1/24 as the default gateway for HostA. The 172.16.31.1/30 address could be the default gateway for SwitchB, and the 172.16.41.1/24 address could be the default gateway for RouterA. The default gateway for HostA must be an address in the same IP network range as HostA. HostA is unable to reach the 172.16.31.0/24 network or the 172.16.41.0/24 network without a static route or a properly assigned default gateway.

Original Exam D Question 47

SwitchA and SwitchB are connected by an 802.1Q trunk link with the default settings.

Which of the following is most likely to occur if you change the native VLAN to VLAN 10 on the trunk interface of SwitchB? (Select the best answer.)

Traffic will be sent between the two switches, but problems could occur because of

oe a native VLAN mismatch.

B. No traffic will be sent between the two switches.

Traffic will be sent between the two switches, and traffic sent over VLAN 10 will be untagged.

Traffic from VLANs other than VLAN 10 will be sent between the two switches, but no traffic from VLAN 10 will be sent between the switches.

Correct Answer: A

Explanation

Traffic will be sent between the two switches, but problems could occur because of a native virtual local area network (VLAN) mismatch. SwitchA is configured to use the default native VLAN, VLAN 1. Modifying the native VLAN to VLAN 10 on SwitchB could cause problems because of the different native VLANs. A mismatched native VLAN configured on either of the two ends of a trunk link could cause problems when traffic is sent by using one of the configured native VLANs. The traffic may be sent, but the native VLAN mismatch could potentially cause the traffic to be misdirected or dropped. In addition, when Dynamic Trunking Protocol (DTP) is used to negotiate the formation of a trunk link between switches, DTP uses the native VLAN for its packets. If the native VLAN is not the same on both ends of the link, a trunk will not dynamically form.

Spanning Tree Protocol (STP) issues, such as unexpected loops, can occur if there is a native VLAN mismatch on the ends of a trunk link. If Per-VLAN Spanning Tree Plus (PVST+) is enabled, one or more of the following error messages might appear on the console to indicate a native VLAN mismatch:

```
%SPANTREE-SP-2-RECV-PVID-ERR: Received BPDU with inconsistent peer Vlan id 1 on
GigabitEthernet1/1 VLAN10
```

```
%SPANTREE-SP-2-BLOCK-PVID-PEER: Blocking GigabitEthernet1/1 on VLAN0001. Inconsistent peer
vlan.
```

```
%SPANTREE-SP-2-BLOCK-PVID-LOCAL: Blocking GigabitEthernet1/1 on VLAN0001. Inconsistent local
vlan.
```

Traffic over the native VLAN is not tagged, which means that an 802.1Q header is not added to the frame. When a switch receives a frame without an 802.1Q header, the switch knows that the frame is part of the native VLAN. Thus both SwitchA and SwitchB should be configured with the same native VLAN in order to ensure that traffic flows correctly between the switches.

Original Exam D Question 68

Which of the following statements are true regarding switches that support a voice VLAN? (Select 2 choices.)

- A. PortFast must be disabled on a port that supports a voice VLAN.
- B. 802.1X authentication can be configured on a port that supports a voice VLAN.
- C. CDP cannot be enabled on a port that supports a voice VLAN.
- D. The voice VLAN feature is disabled by default.

Correct Answer: B, D

Explanation

Of the available choices, the voice virtual local area network (VLAN) feature is disabled by default, and 802.1X authentication can be configured on a port that supports a voice VLAN. You can enable 802.1X authentication on a switch to ensure that a client is authenticated before the client is able to send traffic through the switch. Before the client is authenticated by 802.1X authentication, traffic through the port is restricted to Extensible Authentication Protocol over LAN (EAPOL), Spanning Tree Protocol (STP), and Cisco Discovery Protocol (CDP) traffic. When a client is authenticated, normal traffic is permitted to pass through the port.

To enable the voice VLAN feature, you should issue the switchport voice vlan interface configuration command. Voice VLAN can be configured four different ways:

* You can specify the creation of a unique VLAN to carry voice traffic separate from data traffic by issuing the switchport voice vlan vian-id interface configuration command, where vlan-id is a unique VLAN number that ranges from 1 through 4094. Issuing the switchport voice vlan vian-id command configures the port to carry voice traffic in 802.1Q frames.

* You can issue the switchport voice vlan dot1p interface configuration command to configure the port to carry voice traffic in 802.1p priority-tagged frames. A voice VLAN can be configured to carry voice traffic with data traffic.

* You can issue the switchport voice vlan untagged interface configuration command to enable voice traffic to be carried on the native VLAN.

* You can issue the switchport voice vlan none interface configuration command to enable voice traffic to be carried on the access VLAN, which is the default configuration.

There are several additional configuration requirements that a switch interface must meet to support a voice VLAN. One of these requirements is that CDP must be enabled so that configuration information can be passed between the switch and the Internet Protocol (IP) phone. Typically, CDP is enabled by default on all interfaces. If CDP has been disabled, you can enable it by issuing the edp enable interface configuration command on the port that will support the voice VLAN.

Another feature that must be enabled is the PortFast feature. PortFast, which is enabled by default when a voice VLAN is configured, enables a switch port to bypass the STP listening and learning states and transition immediately into the STP forwarding state. By skipping the listening and learning states, the port has immediate connectivity on the network. If the port were required to cycle through the listening and learning states, the port would be unusable for up to 50 seconds while STP processed any topology changes on the port.

Days 20-21 - Spanning Tree Protocol (STP)

Included Exam D items: Q22, Q23

Original Exam D Question 22

MAC: 0000.0c63.ef20

100 Mbps

MAC: 0000.0c65.4ba6

SwitchD

SwitchE

MAC: 0000.0c60.0717 MAC: 0000.0c65.93ff

You recently became the administrator of the network shown above. You are attempting to determine the roles of each of the switches in the STP topology. You issue the show spanning-tree command on SwitchA and receive the following truncated output:

```
SwitchA#show spanning-tree
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 8192
Address 0000 .0c65.93FF
Cost 23
Port 1 (GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0000.0c63.ef20
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300
<output omitted>

Which of the following switches is the root bridge? (Select the best answer.)

- A. SwitchA
- B. Switchc
- C. SwitchE
- D. SwitchD
- E. SwitchB

Correct Answer: C

Explanation

SwitchE is the root bridge. The show spanning-tree command displays the root ID of the root bridge and the bridge ID (BID) for the current switch. The Media Access Control (MAC) address listed under Root ID is 0000.0c65.93ff, the MAC address for SwitchE; therefore, SwitchE must be the root bridge. Issuing the show spanning-tree command on SwitchE would show that the root ID and BID are the same. In addition, issuing the show spanning-tree detail command on SwitchE would return output containing the line We are the root of the spanning tree.

Electing a root bridge is one of the first steps in the Spanning Tree Protocol (STP) loop-prevention process. STP also optimizes the network by finding the lowest cost paths, determines which port becomes the root port of each switch, and determines which ports become designated ports. STP determines port states through the use of the Spanning Tree Algorithm (STA). The STA uses the following procedure to determine port state:

- © A root switch is elected, and each active port on the root switch is placed into the forwarding state.
- © Each nonroot switch places the port with the lowest-cost path to the root switch, which is known as the root port, into the forwarding state.
- © Ports with the lowest cost to nonroot switches, which are known as designated ports, are also placed into the forwarding state.
- c All other ports are placed into the blocking state.

SwitchA is not the root bridge. If the MAC address listed under Root ID were the same as SwitchA's MAC address listed under Bridge 1D, SwitchA would have been the root bridge. In order for SwitchA to become the root bridge, the priority of SwitchA, which is 32769, would have to be lower than the priority of SwitchE, which is 8192. The bridge priority can be set by issuing the spanning-tree priority value command, where value is a multiple of 4096 in the range from 0 through 61440.

SwitchB, SwitchC, and SwitchD are not the root bridge, because their MAC addresses are not listed as the address for the root ID. In order for any of these switches to become the root bridge, the priority would have to be set to a value lower than 8192.

Original Exam D Question 23

Which of the following statements are true about the PortFast feature? (Select 2 choices.)

- A. PortFast immediately moves the switch port to the STP forwarding state.
- B. PortFast prevents network loops from forming.
- C. PortFast is not automatically disabled when a voice VLAN is disabled.
- D. PortFast can be configured as a global default only.
- E. PortFast should not be enabled for ports that are connected to servers.

Correct Answer: A, C

Explanation

PortFast immediately moves the switch port to the Spanning Tree Protocol (STP) forwarding state, bypassing the listening and learning states. PortFast is a feature offered by Catalyst switches that enables faster connectivity for hosts connected to an access-layer switch port. If PortFast is not enabled, a switch port transitions through the STP listening and learning states before it enters the forwarding state. This process can take as long as 30 seconds if the default STP timers are used. In addition, port initialization could take as long as 50 seconds if Port Aggregation Protocol (PAgP) is enabled. Because host computers or Internet Protocol (IP) phones do not typically perform bridging functions, it is not necessary to make the switch port transition through the normal STP states, because the port should not encounter a bridging loop. Thus STP skips the listening and learning states and places the port into the forwarding state so that the end host has immediate network connectivity.

When a voice virtual local area network (VLAN) is configured, PortFast is automatically enabled; however, PortFast is not automatically disabled when that same voice VLAN is disabled. Typically, voice VLANs are configured on access ports. Likewise, PortFast should also be enabled only on ports that are in the access mode. If PortFast is enabled on a port that is connected to another switch, the potential for creating spanning tree loops significantly increases.

PortFast should be used only on ports that are connected to an end host, such as an IP phone, a client workstation, or a server. Typically, servers and client workstations do not perform bridging duties, so there is no need to delay network connectivity while STP cycles through the listening and learning states. Conversely, PortFast should not be enabled on a port that is connected to a switch or other networking device. If you enable PortFast on such a port, you risk creating bridging loops because the port is permanently in the STP forwarding state.

PortFast can be enabled as a global default as well as on a specific port. If you enable PortFast as a global default, each port that is configured as an access port is enabled with PortFast. You can enable PortFast as a global default by issuing the spanning-tree portfast default global configuration command. You can also enable PortFast on a per-port basis by issuing the spanning-tree portfast interface configuration command.

Day 22 - Rapid STP

Included Exam D items: Q69

Original Exam D Question 69

Which of the following statements is correct regarding RSTP port roles? (Select the best answer.)

- A. Every switch has one root port.
- B. Every nonroot switch has one designated port.
- C. Every switch has one designated port.
- D. Every nonroot switch has one root port.
- E. Every switch has more than one designated port.
- F. Every nonroot switch has more than one root port.

Correct Answer: D

Explanation

Every nonroot switch has one root port. The root port on a switch is the port that receives the best bridge protocol data unit (BPDU), which indicates the best path to the root bridge based on the best root port cost. A root port is always in the forwarding state. Because there is only one best path to the root bridge, a switch cannot have more than one root port. Only the root bridge does not have a root

port.

Switches may have one or more designated ports, and some switches may not have any designated ports. One designated port is selected for each segment. The designated port is the port that is closest to the root bridge based on root port cost. In the event of a tie, the port on the switch with the lowest bridge ID (BID) becomes the designated port. A designated port is normally in the forwarding state. All the ports on a root bridge are designated ports.

The BID consists of a 2-byte bridge priority and a 6-byte Media Access Control (MAC) address. The bridge priority is considered first in the determination of the lowest BID. When two or more switches have the lowest priority, the switch with the lowest MAC address will have the lowest BID; with MAC addresses, numbers are lower than letters and the hexadecimal value A is lower than the hexadecimal value F.

In the following network, all links are running at 100 Mbps, SwitchA is the root bridge, SwitchB has one designated port, and SwitchC has no designated ports:

Priority: 32768
MAC: 0000.0c65.9f20

SwitchC
Priority: 32768
MAC: 0000.0c65.b3a6

SwitchB
Priority: 32768
MAC: 0000.0c65.c292

SwitchA is the root bridge because it has the lowest BID; all the switches have the same priority, and SwitchA has the lowest MAC address. All the ports on SwitchA are designated ports because SwitchA is the root bridge. Port Fa0/0 on SwitchB and port Fa0/0 on SwitchC are root ports because they are closest to the root bridge. Ports Fa0/1 on SwitchB and Fa0/1 on SwitchC advertise the same root port cost because the bandwidth values of the links connected to the root bridge are equal. Therefore, port Fa0/1 on SwitchB is the designated port on the segment between SwitchB and SwitchC because the BID of SwitchB is lower than the BID of SwitchC; both switches have the same priority, and SwitchB has the lower MAC address. SwitchC does not contain any designated ports. Port Fa0/1 on SwitchC is an alternate port, which is a port that leads to the root bridge but is not a root port; an alternate port is always in the blocking state.

Day 23 - EtherChannel

Included Exam D items: Q11

Original Exam D Question 11

=<

You want to establish an EtherChannel between SwitchA and SwitchB that uses an IEEE standards-based protocol.

Which of the following channel-group modes could you configure on the switches? (Select 2 choices.)

- ☐ A. SwitchA set to auto and SwitchB set to auto
- ☐ B. SwitchA set to on and SwitchB set to on
- ☐ C. SwitchA set to passive and SwitchB set to passive
- ☐ D. SwitchA set to active and SwitchB set to active
- ☐ E. Switch set to desirable and SwitchB set to desirable

☐ F. SwitchA set to auto and SwitchB set to desirable

☐ G. SwitchA set to passive and SwitchB set to active

Correct Answer: D, G

Explanation

You could set one switch to active and the other switch to passive; alternatively, you could set both switches to active. Link Aggregation Control Protocol (LACP) is an Institute of Electrical and Electronics Engineers (IEEE) standards-based protocol that groups individual physical ports into a single logical link, called an EtherChannel. The ports that constitute an EtherChannel are grouped according to various parameters, such as hardware, port, and administrative limitations. Because LACP is a standards-based protocol, it can be used between Cisco and non-Cisco switches.

You can issue the `show etherchannel summary` command to verify the status of an EtherChannel link and to determine which aggregation protocol, if any, was used to establish the link. The following sample output indicates that the LACP protocol was used to successfully establish an EtherChannel link:

```
SwitchA#show etherchannel summary
Flags: D - down P - bundled in port-channel
I - stand-alone s - suspended
```

```
- Hot-standby (LACP only)
- Layer3 S - Layer2
in use # - failed to allocate aggregator
```

```
- not in use, minimum links not met
```

```
unsuitable for bundling
```

```
E2 Pw =
```

```
waiting to be aggregated
d - default port
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 2.
```

```
Group Port-channel Protocol _ - Ports
```

```
1 Po1(US) LACP Fae/5(P) Fae/6(P)
```

Port Aggregation Protocol (PAgP) is an older, Cisco-proprietary alternative to LACP. Because PAgP is a Cisco-proprietary protocol, it can be used only on Cisco switches. Like LACP, PAgP identifies neighboring ports and their group capabilities; however, PAgP does not assign roles to the EtherChannel's endpoints like LACP does.

The following table displays the channel-group configurations that will establish an EtherChannel:

```
SwitchB off auto desirable | passive | active on
```

```
SwitchA
```

```
off NO NO NO NO
```

```
auto NO PAgP NO NO
```

```
desirable PAgP PAgP NO NO
```

```
passive NO NO NO LACP
```

active NO NO LACP LACP

on | NO NO NO NO

The channel-group command configures the EtherChannel mode. The syntax of the channel-group command is channel-group number mode {on | active | passive | {auto | desirable} [non-silent]}, where number is the port channel interface number. The on keyword configures the channel group to unconditionally create the channel with no LACP or PAgP negotiation.

The active and passive keywords can be used only with LACP. The active keyword configures the channel group to actively negotiate LACP, and the passive keyword configures the channel group to listen for LACP negotiation to be offered. Either or both sides of the link must be set to active to establish an EtherChannel over LACP; setting both sides to passive will not establish an EtherChannel over LACP.

The auto, desirable, and non-silent keywords can be used only with PAgP. The desirable keyword configures the channel group to actively negotiate PAgP, and the auto keyword configures the channel group to listen for PAgP negotiation to be offered. Either or both sides of the link must be set to desirable to establish an EtherChannel over PAgP; setting both sides to auto will not establish an EtherChannel over PAgP. The optional non-silent keyword requires that a port receive PAgP packets before the port is added to the channel.

Days 24-25 - Dynamic Routing, RIP & EIGRP

Included Exam D items: Q71

Original Exam D Question 71

192.168.0.192/28

192.168.0.128/28

- F

~

RouterA

~

-S~.

RouterB 192.168.0.48/30

'outerC

192.168.0.16/30 192.168.0.36/30

- A

em

RouterD

192.168.0.144/28

192.168.0.160/28

You administer the network shown in the diagram. You issue the show ip route command on RouterD and receive the following output:

RouterD#show ip route

<output omitted>

Gateway of last resort is not set

192.168.0.0/24

c2707nN9 MMO OD

192.
192.
192.
192.
192.
192.
192.
192.
192.
192.

168.
168.
168.
168.
168.
168.
168.
168.
168.
168.

0.128/28 [90/2195456] via 192.168.0.37, 00:
0.160/28 [90/2681856] via 192.168.0.18, 00:03:31, Serial/e
0.48/30 is directly connected, Serial/2

0.16/30 is directly connected, Serial/e

2.
0.
e.
8.
0.

is variably subnetted, 9 subnets, 2 masks

231, Serial/1

36/30 is directly connected, Serial/1

144/28 is directly connected, Fast&thernet0/0

4/30 [90/2195456] via 192.168.0.18, 00:03:31, Serial/e
192/28 [90/2195456] via 192.168.0.18, 00:03:31, Serial/e
-8/30 [90/2681856] via 192.168.0.37, 00:03:31, Serial/1

Which interface and routing protocol on RouterD will be used to forward a packet from the 192.168.0.144/28 network to the 192.168.0.128/28 network? (Select 2 choices.)

A. EIGRP

O 8B. directly connected

Oc. Ospr

D. RIP

E. Serial 0/1

F. Serial 0/0

OG. Serial 0/2

Correct Answer: A, E

Explanation

The Serial 0/1 interface and Enhanced Interior Gateway Routing Protocol (EIGRP) will be used on RouterD to send a packet from the 192.168.0.144/28 network to the 192.168.0.128/28 network. The letter D at the beginning of the routing entries indicates that EIGRP was used to determine the paths to nondirectly connected networks. The show ip route command also displays the administrative distance (AD) and metric, the Internet Protocol (IP) address of the next hop, and the outgoing interface for each route.

When a packet is sent to a router, the router checks the routing table to see whether the next-hop address for the destination network is known. Examination of the show ip route output shows that nine different routes have been added to RouterD's routing table. In this scenario, a packet sent from the 192.168.0.144/28 network to the 192.168.0.128/28 network will follow the path from RouterD to RouterC and then on to RouterA, where it will be forwarded to the 192.168.0.128 network. Although the topology pictured above does not indicate the location of the Serial 0/1 interface on RouterD, the displayed routing table indicates that the Serial 0/1 interface has an address of 192.168.0.37 with a 30-bit subnet mask. This particular IP address belongs to the 192.168.0.36/30 network, which connects RouterD to RouterC.

When the show ip route command is issued, the initial output includes a key that can be used to determine the routing protocols used to learn routes to other networks. The display key looks like this:

RouterA#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

- candidate default, U - per-user static route, o - ODR

ry

P - periodic downloaded static route

<output omitted>

The letter D is used to specify a route learned through the use of EIGRP. Other codes displayed in this key include the letter O for Open Shortest Path First (OSPF) and the letter R for Routing Information Protocol (RIP). Directly connected routes are denoted in the routing table by the letter c. The routing table above indicates that neither OSPF nor RIP has been configured for this topology.

The Serial 0/0 interface would not be used to send a packet from the 192.168.0.144/28 network to the 192.168.0.128/28 network. By examining the output from the show ip route command, you can see that the Serial 0/0 interface has an IP address of 192.168.0.18. This address is part of the 192.168.0.16/30 subnet, which is the network used to connect RouterD to RouterB. However, RouterD's routing table displays a route to the 192.168.0.128/28 network via the Serial 0/1 interface.

The Serial 0/1 interface has an IP address of 192.168.0.37, which belongs to the 192.168.0.36/30 subnet and is used to connect RouterD to RouterC.

The Serial 0/2 interface would not be used to send a packet from the 192.168.0.144/28 network to the 192.168.0.128/28 network. By examining the output from the show ip route command, you can see

that the Serial 0/2 interface is used only to connect to the directly connected 192.168.0.48/30 network.

Days 26-28 - OSPF

Included Exam D items: Q1, Q20, Q41, Q43, Q53

Original Exam D Question 1

You are a junior network administrator and have been asked by your supervisor to complete some network configuration tasks that were partially completed by another network administrator during a prior shift. The routers in your network have been partially configured with IPv4 addresses and an OSPF process has been partially configured on each router; however, several interfaces still require IPv4 address assignments and these interfaces must be added to OSPF Area 0.

You have been given the following tasks to complete the configuration of your organization's network:

Configure the FastEthernet 0/1 interface on RouterA to reside in a subnet of the 172.16.1.0 network. The subnet should be as small as possible while ensuring that at least 50 host addresses are available. The FastEthernet 0/1 interface should be assigned the last usable IP address in the subnet.

Configure RouterA to advertise its newly configured subnet as part of OSPF Area 0 by specifying the area in interface configuration mode.

Configure the FastEthernet 0/1 interface on RouterB to reside in a subnet of the 172.16.2.0 network. The subnet should be as small as possible while ensuring that at least five host addresses are available. The FastEthernet 0/1 interface should be assigned the last usable IP address in the subnet.

Configure Router to advertise its newly configured subnet as part of OSPF Area 0 by specifying the area in interface configuration mode.

Configure the FastEthernet 0/1 interface on RouterC to reside in a subnet of the 172.16.3.0 network. The subnet should be as small as possible while ensuring that at least 15 host addresses are available. The FastEthernet 0/1 interface should be assigned the last usable IP address in the subnet.

Configure RouterC to advertise its newly configured subnet as part of OSPF Area 0 by specifying the area in interface configuration mode.

Configure a router ID of 9.9.9.9 on the appropriate router to ensure that RouterA will become the BDR for the 198.51.100.0/24 subnet if either the current BDR or DR should happen to fail.

Click the routers in the topology to access their consoles and complete the configurations. You should make no changes other than the ones required to accomplish the tasks. You should not make configuration changes to SwitchA or to the three workstations: HostA, HostB, and HostC. Ensure that you save your configuration changes on each router before ending the simulation.

Lab task - no separate correct-answer section in the source PDF.

Original Exam D Question 20

An OSPF router is attempting to obtain its router ID.

Which of the following will take precedence? (Select the best answer.)

A. manually configured router ID

- B. the highest IP address on a loopback interface
- C. the lowest IP address on a nonloopback interface
- D. the lowest IP address on a loopback interface
- E. the highest IP address on a nonloopback interface

Correct Answer: A

Explanation

Of the choices provided, a manually configured router ID will take precedence over any form of automatically calculated router ID on an Open Shortest Path First (OSPF) router. OSPF uses router IDs in the designated router (DR) and backup designated router (BDR) election process.

If the OSPF router ID of a router is not manually configured, the router ID will be determined first by the highest Internet Protocol (IP) address among loopback interfaces and then by the highest IP address among physical interfaces if no loopback interfaces have been configured. Because loopback interfaces are logical, not physical, a loopback interface can never fail. For this reason, configuring a loopback interface is recommended for OSPF operation, because loopback interfaces are more reliable than physical interfaces.

An OSPF router does not consider the lowest IP address of any interface when it attempts to obtain a router ID. However, a switch that is using Spanning Tree Protocol (STP) will use the lowest bridge ID (BID) when it attempts to determine the root bridge. The BID is composed of a 2-byte bridge priority and a 6-byte Media Access Control (MAC) address. The bridge priority is considered first in the determination of the lowest BID. When two or more switches have the lowest priority, the switch with the lowest MAC address becomes the root bridge.

Original Exam D Question 41

You are attempting to configure OSPF between RouterA and RouterB on your network. However, the routers are unable to form an OSPF adjacency. You issue the `show ip ospf interface fastethernet 0/0` command on each router and receive the following output:

```
RouterA#show ip ospf interface fastethernet 0/0
FastEthernet 0/0 is up, line protocol is up
Internet Address 10.1.1.13/30, Area 0
Process ID 101, Router ID 192.168.15.237, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.15.237, Interface address 10.1.1.13
No Backup Designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:07
Index 1/1, flood queue length 0
Next 0x0(0)/2xe(e)
Last flood scan length is 2, maximum is 2
Last flood scan time is 0 msec, maximum is 4 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
```

```
RouterB#show ip ospf interface fastethernet 0/0
FastEthernet 0/0 is up, line protocol is up
Internet Address 10.1.1.14/30, Area 0
Process ID 102, Router ID 192.168.15.237, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.15.237, Interface address 10.1.1.14
No Backup Designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:05
Index 1/1, flood queue length 0
```

Next x0(®)/0xe(e)

Last flood scan length is 2, maximum is 2

Last flood scan time is 0 msec, maximum is 4 msec

Neighbor Count is 0, Adjacent neighbor count is 0

Suppress hello for 0 neighbor(s)

Why are the routers unable to form an adjacency? (Select the best answer.)

- A. The IP addresses are configured with the wrong subnet mask.
- B. The router IDs must not match.
- C. The dead timer value must match the hello timer value.
- D. The OSPF areas must match.
- E. The cost must be higher than 1.
- F. The OSPF process IDs must match.

Correct Answer: B

Explanation

The routers are unable to form an adjacency because the router IDs must not match. Router IDs must be unique; a router will ignore hello packets that are marked with its own router ID. To manually configure the router ID, you should issue the router-id ip-address command in router configuration mode, where ip-address is a 32-bit value in dotted decimal notation. If the router ID is not manually configured, the router ID will be the highest loopback Internet Protocol (IP) address configured on a router. If a loopback IP address is not configured, then the router ID will be the highest IP address among the configured interfaces on the router.

The Open Shortest Path First (OSPF) areas in this scenario match. Only OSPF routers in the same hierarchical area form adjacencies. To configure an OSPF area, you should issue the

network address mask area area command in OSPF router configuration mode. Although you can configure a multiarea OSPF topology, the OSPF areas on neighboring interfaces must match in order for two OSPF routers to form an adjacency. In this scenario, RouterB's FastEthernet 0/0 interface is operating in Area 0 and is directly connected to RouterA's FastEthernet 0/0 interface, which is also operating in Area 0. Area 0 is also known as the backbone area. If you are not using virtual links, any nonbackbone OSPF areas you configure must border Area 0. On smaller networks, single-area OSPF configurations also offer the benefit of generating fewer link-state advertisements (LSAs).

Whereas Enhanced Interior Gateway Routing Protocol (EIGRP) process IDs must match when EIGRP is used, OSPF process IDs do not have to match when OSPF is used. The OSPF process ID is locally significant to the router and can be any positive integer in the range from 1 through 65535. You can specify the OSPF process ID by issuing the router ospf process-id command when you configure a router for OSPF.

The cost does not have to be higher than 1. By default, OSPF uses the bandwidth of a link to determine the cost. The higher the bandwidth, the lower the cost. To calculate the cost, divide 100,000,000 by the bandwidth in bits per second (bps). The FastEthernet 100-megabits per second (Mbps) link between RouterA and RouterB would have a default cost of 1, because $100,000,000 \div 100,000,000 = 1$. To manually configure the OSPF cost of a link, you should issue the ip ospf cost cost command in interface configuration mode.

The IP addresses for RouterA and RouterB are within the same subnet. Therefore, the subnet mask is correct, and the two routers should be able to communicate. A /30 subnet mask allows for two hosts per subnet. In this example, 10.1.1.12/30 is the network address, 10.1.1.13/30 and 10.1.1.14/30 are valid host addresses, and 10.1.1.15/30 is the broadcast address.

The dead timer value should not match the hello timer value. In order for OSPF routers to establish an adjacency, the dead timer on one router should match the dead timer on the other router and the hello timer on one router should match the hello timer on the other router. Both RouterA and

RouterB are set to a hello interval of 10 seconds and a dead interval of 40 seconds, so the OSPF timers are configured correctly. The hello timer is used to specify the amount of time between sending hello packets, and the dead timer is used to specify the amount of time to wait for hello packets before declaring a neighbor to be down. By default, the hello timer is set to 10 seconds on point-to-point and broadcast links and 30 seconds on nonbroadcast multiaccess (NBMA) links. The dead timer is set to four times the hello timer value by default. To manually configure the hello timer interval, you should issue the `ip ospf hello-interval seconds` command in interface configuration mode. To manually configure the dead timer interval, you should issue the `ip ospf dead-`

`interval seconds` command in interface configuration mode.

Original Exam D Question 43

)

a

-

outerA

You administer the OSPF network above.

You issue the `default-information originate` command on RouterA.

Which of the following statements are true? (Select 2 choices.)

A. RouterA will become the OSPF ASBR.

OD 8B. OSPF will redistribute all of RouterA's directly connected routes.

C. OSPF will advertise RouterA's gateway of last resort.

D. RouterA will become the OSPF ABR.

E. OSPF will summarize all of RouterA's directly connected routes.

Correct Answer: A, C

Explanation

Open Shortest Path First (OSPF) will advertise RouterA's gateway of last resort and RouterA will become an autonomous system boundary router (ASBR) if you issue the `default-information originate` command on RouterA. The `default-information originate` command configures OSPF to inject a router's default external route into OSPF. In addition, if you issue either the `default-information originate` command or a `redistribute network` command in OSPF router configuration mode, Cisco IOS automatically considers the router an ASBR. An ASBR redistributes routes from other routing protocols into the OSPF domain.

If RouterA in this scenario were configured with the `ip route 0.0.0.0 0.0.0.0 FastEthernet0/1` command, any traffic destined for a network that is not in RouterA's routing table would be forwarded through the FastEthernet 0/1 interface on RouterA. However, if OSPF does not redistribute or advertise the default route on RouterA, neither RouterB nor RouterC will be capable of forwarding traffic to a destination address that does not have an associated path in their routing tables.

If you issue the `default-information originate` command on RouterA in this scenario, RouterA will become RouterB's gateway of last resort. Similarly, RouterB will become RouterC's gateway of last resort because the path to RouterA from RouterC is through RouterB.

OSPF will not redistribute all of RouterA's directly connected routes. The `default-information originate` command causes OSPF to advertise the gateway of last resort, which might be a directly connected route. However, it does not cause OSPF to advertise all directly connected routes. If you wanted to configure OSPF on RouterA to redistribute all of RouterA's connected routes, you would issue the `redistribute connected` command in OSPF router configuration mode.

OSPF will not summarize all of RouterA's directly connected routes. The `summary-`

address address mask command summarizes routes that have been redistributed into OSPF. This command should be issued on an ASBR, not an area border router (ABR). To summarize routes on an ABR, you should issue the area area-id range address mask command.

RouterA will not become an ABR. When the default-information originate command is issued on an OSPF router, the router automatically becomes an ASBR.

Original Exam D Question 53

You configure OSPF on a router and advertise all available interfaces. The Gigabit Ethernet interfaces are using the default OSPF network type and timer values.

Which of the following statements is correct regarding how OSPF operates on the Gigabit Ethernet interfaces? (Select the best answer.)

- A. The hello timer is set to 30 seconds, and the dead timer is set to 120 seconds.
- B. Unicast updates are sent.
- C. DR and BDR elections are performed.
- D. The neighbor command is required to establish adjacencies.

Correct Answer: C

Explanation

Designated router (DR) and backup designated router (BDR) elections are performed. The Gigabit Ethernet interfaces use the Open Shortest Path First (OSPF) broadcast network type by default. There are five OSPF network types:

- © Broadcast
- © Nonbroadcast
- © Point-to-point
- © Point-to-multipoint broadcast
- © Point-to-multipoint nonbroadcast

On broadcast networks, multicast updates are sent, so manual configuration of neighbor routers with the neighbor command is not required. By default, the hello timer is set to 10 seconds and the dead timer is set to 40 seconds. To configure an OSPF broadcast network, you should issue the ip ospf network broadcast command. The OSPF broadcast network type is enabled by default on Fiber Distributed Data Interface (FDDI) and Ethernet interfaces, including Fast Ethernet and Gigabit Ethernet interfaces.

On nonbroadcast networks, DR and BDR elections are performed. Nonbroadcast networks do not allow multicasts; therefore, manual configuration of neighbor routers with the neighbor command is required so that OSPF sends unicast updates. By default, the hello timer is set to 30 seconds and the dead timer is set to 120 seconds. To configure an OSPF nonbroadcast network, which is also called a nonbroadcast multiaccess (NBMA) network, you should issue the ip ospf network non-broadcast command. The OSPF nonbroadcast network type is enabled by default on Frame Relay and X.25 interfaces.

On point-to-point networks, DR and BDR elections are not performed. Multicast updates are sent, so manual configuration of neighbor routers with the neighbor command is not required. By default, the hello timer is set to 10 seconds and the dead timer is set to 40 seconds. To configure an OSPF point-to-point network, you should issue the ip ospf network point-to-point command. The OSPF point-to-point network type is enabled by default on High-Level Data Link Control (HDLC) and Point-to-Point Protocol (PPP) serial interfaces.

OSPF point-to-multipoint broadcast networks operate just like OSPF point-to-point networks except the hello timer is set to 30 seconds and the dead timer is set to 120 seconds by default. To configure an OSPF point-to-multipoint broadcast network, you should issue the `ip ospf network point-to-multipoint` command.

OSPF point-to-multipoint nonbroadcast networks operate just like OSPF point-to-multipoint broadcast networks except that multicasts cannot be sent; therefore, manual configuration of neighbor routers with the `neighbor` command is required so that OSPF sends unicast updates. To configure an OSPF point-to-multipoint nonbroadcast network, you should issue the `ip ospf network point-to-multipoint non-broadcast` command.

Day 29 - First Hop Redundancy Protocols

Included Exam D items: Q8

Original Exam D Question 8

Which of the following HSRP routers has the highest priority? (Select the best answer.)

- A. the active router
- B. the backup router
- C. the master router
- D. the standby router

Correct Answer: A

Explanation

The Hot Standby Router Protocol (HSRP) active router has the highest priority. When HSRP is configured, routers are placed into an HSRP group. The router with the highest priority value, or the router with the highest Internet Protocol (IP) address if more than one router has the highest priority value, is elected as the active router for the group. The router with the next-highest priority is elected as the standby router. All remaining routers in the group are placed in the listen state. The active router is responsible for forwarding traffic sent to the gateway IP address. If the active router fails or becomes unavailable, the standby router will assume the role of the active router and another router will be elected as the standby router.

Virtual Router Redundancy Protocol (VRRP) has one master router and one or more backup routers. The VRRP master router has the highest priority. By default, the VRRP priority is set to 100; the VRRP priority can be a value from 1 through 254. Routers are assigned to a VRRP group, and the group functions as a single gateway for clients. A VRRP group has one master router; all other routers in the VRRP group are backup routers. The master router is used to forward packets to different virtual local area networks (VLANs). When VRRP is implemented, the group of devices is assigned a virtual IP address and a virtual Media Access Control (MAC) address. Clients that utilize a VRRP platform send packets to the virtual IP address. These packets are received and forwarded from the current master router. Backup routers receive advertisements from the master router that inform the backup routers of the master router's state. These advertisements are sent every second by default. If updates are not received for a specified amount of time, the master device is assumed to have failed and the election of a new master device begins.

Day 30 - TCP & UDP

Included Exam D items: Q28, Q65

Original Exam D Question 28

Which of the following Application layer protocols use UDP for unsynchronized, connectionless data transfer? (Select 2 choices.)

OA HTTP
B. TFIP
Oc. Fp

O pb. smtp

O & SNMP

Correct Answer: B, E

Explanation

Simple Network Management Protocol (SNMP) and Trivial File Transfer Protocol (TFTP) use User Datagram Protocol (UDP) for unsynchronized, connectionless data transfer. UDP is a Transport layer protocol that does not use sequence numbers or establish synchronized connections. Because of UDP's connectionless nature, transmitted datagrams can appear out of sequence or can be dropped without notice; thus it is the responsibility of the Application layer protocol to reorder packets or request the transmission of lost datagrams. SNMP is used to monitor and manage network devices. TFTP uses UDP port 69 to transfer files unreliably and without authentication over a network. Administrators can use TFTP to transfer Cisco IOS images from a server to a device to perform firmware upgrades. Other common Application layer protocols that use UDP include Dynamic Host Configuration Protocol (DHCP), which is used to assign Internet Protocol (IP) addressing information to clients, Network Time Protocol (NTP), which is used to coordinate time on a network, and Remote Authentication Dial-In User Service (RADIUS), which is used to authenticate users.

Hypertext Transfer Protocol (HTTP) and File Transfer Protocol (FTP) use Transmission Control Protocol (TCP) for reliable, connection-oriented data transfer. TCP is a Transport layer protocol that uses sequencing and error-checking to ensure that transmitted data can be easily reordered if packets arrive out of sequence and can be retransmitted if any packets are lost. Because TCP handles data sequencing and the retransmission of lost data, the Application layer protocols that rely on TCP do not need to handle those tasks and can rely on receiving reliable, ordered data. FTP, which is used to transfer files over a network, uses TCP ports 20 and 21 for data and control traffic, respectively. Cisco devices can reliably transfer IOS images by using FTP. FTP requires the transmission of authentication credentials, even if anonymous FTP is in use, but those credentials are transmitted in plain text. Other common TCP protocols are HTTP, which is used to transfer webpages over the Internet; Simple Mail Transfer Protocol (SMTP), which is used to send email messages; Post Office Protocol 3 (POP3), which is used to retrieve email messages; and Telnet, which is used to manage network devices.

Original Exam D Question 65

Which of the following ports should you block if you want to prohibit Telnet traffic? (Select the best answer)

A. 80

B. 53

Oc 110

D. 23

Oe 21

F. 25

Correct Answer: D

Explanation

You should block Transmission Control Protocol (TCP) port 23 if you want to prohibit Telnet traffic. Telnet is used to create a terminal connection to remote devices.

You should block TCP port 21 if you want to prohibit File Transfer Protocol (FTP) traffic. FTP, which is used to transfer files over a network, uses TCP ports 20 and 21.

You should block TCP port 25 if you want to prohibit Simple Mail Transfer Protocol (SMTP) traffic. SMTP is used to send email messages.

You should block TCP and User Datagram Protocol (UDP) port 53 if you want to prohibit Domain Name System (DNS) traffic. DNS is used to translate host names to Internet Protocol (IP) addresses.

You should block TCP port 80 if you want to prohibit Hypertext Transfer Protocol (HTTP) traffic. HTTP is used to transfer webpages over the Internet.

You should block TCP port 110 if you want to prohibit Post Office Protocol 3 (POP3) traffic. POP3 is used to receive email messages.

Days 31-33 - IPv6

Included Exam D items: Q19, Q38, Q42, Q70

Original Exam D Question 19

You issue the following commands on a Cisco router's FastEthernet 0/0 interface:

```
ipv6 address 2001:DB8:A::/64 eui-64  
no shutdown
```

The interface on the other side of the link is not yet configured. In addition, there is no DHCPV6 server on the network.

How many IPv6 addresses are configured on the interface? (Select the best answer.)

- A. one
- B. none
- C. three
- D. two

Correct Answer: D

Explanation

Two Internet Protocol version 6 (IPv6) addresses are configured on the interface after you issue the `ipv6 address 2001:DB8:A::/64 eui-64` command and the `no shutdown` command on the Cisco router's FastEthernet 0/0 interface. The `eui-64` keyword configures a static IPv6 prefix but allows the router to automatically generate a 64-bit interface ID known as an extended unique identifier (EUI)-64 interface ID; the EUI-64 interface ID is based on the interface's Media Access Control (MAC) address.

In addition, a second IPv6 address, referred to as a link-local unicast address, is automatically assigned to an interface when IPv6 is enabled on that interface. Therefore, the IPv6 interface in this scenario is configured with two IPv6 addresses. There are three ways to enable IPv6 on an interface: by manually assigning an IPv6 address to the interface, by automatically assigning an IPv6 address to the interface, or by issuing the `ipv6 enable` command on the interface.

To automatically assign an IPv6 address to an interface, you can implement Stateless Address Automatic Configuration (SLAAC), Dynamic Host Configuration Protocol version 6 (DHCPv6), or a combination of the two. SLAAC configurations occur based on information that is sent in router advertisements from an IPv6 gateway operating on the same network segment. When the link-local interface is active on the segment, it can announce itself and receive router advertisements from an IPv6 router that is operating on the same segment. If an IPv6 router exists and can function as an IPv6 gateway, it will advertise that functionality as well as the globally unique prefix with which it is configured and with which connected nodes should be used. The `ipv6 address autoconfig` command configures an interface to automatically assign itself a global unicast IPv6 address by using

SLAAC.

The same ipv6 address autoconfig command that enables SLAAC on an interface will enable the interface to obtain additional information from a DHCPv6 server if a DHCPv6 server exists on the network and is configured to send nonaddress information. The ipv6 address dhcp command configures a DHCPv6 client interface to use stateful DHCPv6 addressing, which configures addressing information and extra information from the DHCPv6 server.

IPv6 link-local unicast addresses are used for communication over a single link. Routers do not forward traffic sent to a link-local address; the traffic stays on the local link. IPv6 link-local unicast addresses are often used for neighbor discovery. These addresses usually begin with FE8, as specified in Request for Comments (RFC) 4291.

Unlike with IP version 4 (IPv4), it is possible to configure more than one IPv6 address on an interface without defining the addresses as primary or secondary. IPv4-only interfaces can be configured with only one primary IPv4 address. Although it is possible for an interface to have more than two IPv6 addresses, that is not the case in this scenario.

Original Exam D Question 38

Which of the following prefixes will an IPv6-enabled computer use to automatically configure an IPv6 address for itself? (Select the best answer.)

- A. FF00:/8
- B. FE80:/10

- C. 2000:/3
- D. FC00:/7

Correct Answer: B

Explanation

An Internet Protocol version 6 (IPv6)-enabled computer will use the prefix FE80::/10 to automatically configure an IPv6 address for itself. The IPv6 prefix FE80::/10 is used for unicast link-local addresses. IPv6 addresses in the FE80::/10 range begin with the characters FE80 through FEBF. Unicast packets are used for one-to-one communication. Link-local addresses are unique only on the local segment. Therefore, link-local addresses are not routable. Unicast link-local addresses are used for neighbor discovery and for environments in which no router is present to provide a routable IPv6 prefix.

IPv6 was developed to address the lack of available address space with IP version 4 (IPv4). An IPv6 address is a 128-bit (16-byte) address that is typically written as eight groups of four hexadecimal characters, including numbers from 0 through 9 and letters from A through F. Each group of four characters is separated by colons. Leading zeroes in each group can be dropped. A double colon can be used at the beginning, middle, or end of an IPv6 address in place of one or more contiguous four-character groups consisting of all zeroes. However, only one double colon can be used in an IPv6 address. Therefore, the following IPv6 addresses are equivalent:

```
FE80:0000:0000:070D:0000:50A0:0001:0024
FE80::070D:0000:50A0:0001:0024
```

```
c FE80:0:0:70D:0:50A0:1:24
=FE80::70D:0:50A0:1:24
```

An IPv6-enabled computer will not use the prefix 2000::/3 to automatically configure an IPv6 address for itself. The IPv6 prefix 2000::/3 is used for global aggregatable unicast addresses. IPv6 addresses in the 2000::/3 range begin with the characters 2000 through 3FFF. Global aggregatable unicast address prefixes are distributed by the Internet Assigned Numbers Authority (IANA) and are globally routable over the Internet.

An IPv6-enabled computer will not use the prefix FC00::/7 to automatically configure an IPv6 address for itself. The IPv6 prefix FC00::/7 is used for unicast unique-local addresses. IPv6 addresses in this range begin with the characters FCO0 through FDFF. Unique-local addresses are not globally routable, but they are routable within an organization.

An IPv6-enabled computer will not use the prefix FFO0::/8 to automatically configure an IPv6 address for itself. The IPv6 prefix FFO0::/8 is used for multicast addresses, which are used for one-to-many communication. IPv6 addresses in the FF00::/8 range begin with the characters FFO0 through FFFF. However, certain address ranges are used to indicate the scope of the multicast address. The following IPv6 multicast scopes are defined:

© FFO1::/16 - node-local
0 FFO2::/16 - link-local
© FFOS::/16 - unique-local

© FFO8::/16 - organization-local
© FFOE::/16 - global

Original Exam D Question 42

Which of the following statements about unique local unicast IPv6 addresses are true? (Select 2 choices.)

- A. They are unique only within an organization.
- B. They are assigned by ICANN.
- C. They are equivalent to IPv4 multicast addresses.
- D. The first 7 bits of the prefix are always 1111110.
- E. They can communicate only over a single link.

Correct Answer: A, D

Explanation

Internet Protocol version 6 (IPv6) unique local unicast addresses are unique only within an organization. They are similar to IP version 4 (IPv4) Request for Comments (RFC) 1918 addresses in that they are assigned by a local administrator and must be unique only within an organization. These addresses always begin with FC or FD because the first 7 bits of an IPv6 unique local unicast address are always 1111110. Unique local unicast addresses require a randomly generated prefix to ensure that they are unique. Because of the random nature of the network prefix, unique local unicast addresses are not aggregatable and cannot be summarized.

IPv6 global unicast addresses, not unique local addresses, are assigned by the Internet Corporation for Assigned Names and Numbers (ICANN). The IPv6 prefix of 2000::/3 is the global unicast IPv6 prefix. IPv6 global unicast addresses are similar to IPv4 global unicast addresses in that they are globally routable. These addresses are assigned by ICANN to the Regional Internet Registries (RIRs), which distribute the addresses to Internet service providers (ISPs). The ISPs then distribute address ranges to organizations. IPv6 global unicast addresses always begin with a 2 or a 3 because the first 3 bits of an IPv6 global unicast address are always 001.

IPv6 multicast addresses are similar to IPv4 multicast addresses. The IPv6 prefix FF00::/8 is used for multicast addresses, which are used for one-to-many communication. IPv6 addresses in the FF00::/8 range begin with the characters FFO0 through FFFF. However, certain address ranges are used to indicate the scope of the multicast address. The following IPv6 multicast scopes are defined:

e FFO1::/16 - node-local

c FFO2::/16 - link-local

6 - site-local

6 - organization-local

c FFOE::/16 - global

These addresses always begin with FF because the first 8 bits of an IPv6 multicast address are always 11111111.

IPv6 link-local unicast addresses, not unique local addresses, are used for communication over a single link. Routers do not forward traffic sent to a link-local address; the traffic stays on the local link. IPv6 link-local unicast addresses are often used for neighbor discovery. These addresses usually begin with FE8, as specified in RFC 4291. Technically, these addresses could begin with FE9, FEA, or FEB because there are four possible combinations of the first 12 bits of the address. The first 10 bits of an IPv6 link-local unicast address are always 1111111010, or FE, which means that link-local IPv6 addresses could technically begin with any of the following:

- © 1111 1110 1000, which is equal to FES
- © 1111 1110 1001, which is equal to FE9
- © 1111 1110 1010, which is equal to FEA
- © 11111110 1011, which is equal to FEB

Original Exam D Question 70

2001:0B8:1::/64

Bs m2 n

You have connected the GigabitEthernet 0/1 interface of RouterC to RouterB and configured the appropriate interface addresses on both routers.

Which of the following commands could you issue on RouterC to enable communication between RouterC and RouterA? (Select 2 choices.)

- A. ipv6 route 2001:DB8:1::/64 2001:DB8:2::1
- O 8B. ipv6 route 2001:DB8:2::/64 2001:DB8:1::2
- C. ipv6 route 2001:DB8:2::1 2001:DB8:1::/64
- D. ipv6 route ::0 GigabitEthernet 0/1
- E. ipv6 route 2001:DB8:1::2 2001:DB8:2::/64

Correct Answer: A, D

Explanation

You could issue either the ipv6 route ::0 GigabitEthernet 0/1 command or the ipv6 route 2001:DB8:1::/64 2001:DB8:2::1 command on RouterC to enable communication between RouterC and RouterA. However, RouterA also needs to be configured with a route to RouterC before communication will occur.

The basic syntax of the ipv6 route command is ipv6 route prefix/length {interface | next-hop} [administrative-distance], where prefix/length is the destination network, interface is the output interface on the local router, next-hop is the next-hop Internet Protocol version 6 (IPv6) address, and administrative-distance is an optional administrative distance (AD).

You should not issue the ipv6 route 2001:DB8:2::/64 2001:DB8: command on RouterC. This command should be issued on RouterA to create a static route to RouterC. Alternatively, you can

create a default gateway on RouterA by issuing the ipv6 route ::0 GigabitEthernet 0/1 command.

You should not issue the ipv6 route 2001:DB8:2::1 2001:DB8:1::/64 command or the ipv6 route 2001:DB8: 2001:DB8:2::/64 command. The destination network and next-hop IPv6 addresses in these commands are swapped and are therefore invalid.

Days 34-35 - Access Control Lists (ACLs)

Included Exam D items: Q30, Q44, Q54

Original Exam D Question 30

You have issued the following commands on a Cisco router:

```
RouterA(config)#ip access-list extended 111
RouterA(config-ext-nacl)#permit tcp any any eq 443
RouterA(config-ext-nacl)#interface fastethernet 0/0
RouterA(config-if)#ip access-group 111 in
```

Which of the following statements is true regarding inbound traffic on the FastEthernet 0/0 interface? (Select the best answer)

- A. All HTTPS traffic is denied.
- B. All HTTP traffic is denied.
- C. Neither HTTP nor HTTPS traffic is denied.
- D. Both HTTP and HTTPS traffic are denied.

Correct Answer: B

Explanation

In this scenario, all inbound Hypertext Transfer Protocol (HTTP) traffic is denied on the FastEthernet 0/0 interface. Access control lists (ACLs) are used to identify traffic so that matching traffic can then be filtered, analyzed, forwarded, or influenced in various ways. ACLs can be identified by an access list number or an access list name. Numbered access lists ranging from 1 through 99 are standard access lists and can identify traffic based on only the source Internet Protocol (IP) address. Numbered access lists ranging from 100 through 199 and from 2000 through 2699 are extended access lists, and they can identify traffic based on source and destination IP addresses and by traffic type, protocol, and port number.

An extended ACL can be configured by creating an ACL entry sequence that uses the following syntax:

```
access-list access-list-number [dynamic dynamic-name [timeout minutes] {deny | permit}
protocol source source-wildcard [operator [port]] destination destination-wildcard [operator [port]]
[established] [precedence precedence] [tos tos] [log | log-input] [time-range time-range-name]
```

The permit and deny keywords are used to indicate whether matching packets should be forwarded or dropped, respectively. If the packet does not match any of the access list statements, the packet will be dropped; this is called the implicit deny rule.

In this scenario, the permit tcp any any eq 443 command creates an ACL entry sequence that permits Transmission Control Protocol (TCP) traffic from any source address to port 443 on any destination address. TCP port 443 is the default port used for HTTPS traffic. However, because there are no additional ACL entry sequences, all other traffic will match the implicit deny statement at the end of the ACL and will be denied. Therefore, HTTP traffic, which uses TCP port 80 by default, will be denied.

You can add an ACL entry sequence that explicitly permits HTTP traffic by entering named ACL configuration mode and then adding an additional ACL entry sequence that will permit HTTP traffic. An ACL sequence number can be any integer value ranging from 1 through 2147483647. By default, access list entries are assigned sequence numbers in increments of 10. You can add, insert, or delete access list entries in a Cisco named ACL. If an access list statement is added by using the access-list command, the statement will be assigned the next available sequence number based on the last configured sequence number in the ACL. If an access list statement is added from named ACL configuration mode, the access list statement can be assigned any unused sequence number in the

ACL. By assigning a sequence number to an access list statement, you can control its position within the existing ACL. For example, you could insert an access list statement between two existing statements by using a sequence number that is higher than that of the first statement and lower than that of the second statement.

You can place the router into named ACL configuration mode by issuing the `ip access-list`

`extended acl-name` command, where `acl-name` is the name or number you have assigned to the ACL. For example, the following block of commands could be used to add an ACL entry sequence that explicitly permits HTTP traffic by issuing the following block of commands:

```
RouterA(config)#ip access-list extended 111
```

```
RouterA(config-ext-nacl)#permit tcp any any eq www
```

The `permit tcp any any eq www` commands creates a new ACL entry sequence that permits HTTP traffic. Because a sequence number is not specified, the new ACL entry sequence is simply assigned the next available sequence number and appended to the existing ACL. Alternatively, you could issue the `permit ip any any` command to create an ACL control entry to permit all traffic that has not been explicitly filtered by prior access list statements.

Original Exam D Question 44

You issue the following commands on a Cisco router named RouterA:

```
RouterA(config)#ip access-list extended boson
```

```
RouterA(config-ext-nacl)#permit tcp any any range 22 443
```

```
RouterA(config-ext-nacl)#deny tcp any any neq 23
```

```
RouterA(config-ext-nacl)#permit tcp any any eq 20 21
```

```
RouterA(config-ext-nacl)#permit tcp any any lt 442
```

```
RouterA(config-ext-nacl)#permit tcp any any gt 444
```

Which of the following statements about the ACL is true? (Select the best answer.)

A. Traffic that matches TCP destination port 444 will be permitted.

B. Traffic that matches TCP destination ports in the range from 22 through 80 will be permitted.

C. Traffic that matches TCP destination port 20 or TCP destination port 21 will be permitted.

D. Traffic that matches TCP destination port 23 will not be permitted.

Correct Answer: B

Explanation

Traffic that matches Transmission Control Protocol (TCP) destination ports in the range from 22 through 80 will be permitted in this scenario because there are no prior access control list (ACL) rules that prevent the ACL from matching traffic to the `permit tcp any any range 22 443` rule. To create a named extended ACL, you should issue the `ip access-list extended acl-name` command. Issuing this command will place the router in extended-named-ACL mode. When a packet matches an ACL rule, the ACL will not attempt to match the traffic to any subsequent rules that have been configured in the ACL. Extended ACL entries can be created by using the following basic syntax:

```
[sequence-number] {deny | permit} protocol source source-wildcard [operator port [port]  
destination destination-wildcard [operator port [port]]
```

The optional sequence-number value enables an administrator to specify a sequence number to insert a rule into a specific order in an existing ACL. If a value is not specified for sequence number, the ACL will automatically add a sequence number in an interval of 10. In addition, port numbers that map to

ACL keywords will be converted to those keywords, as shown in the following output:

```
RouterA#show ip access-lists
```

```
Extended IP access list boson
10 permit tcp any any range 22 443
20 deny tcp any any neq telnet
30 permit tcp any any eq ftp-data ftp
48 permit tcp any any lt 442
50 permit tcp any any gt 444
```

In the output above, the first rule has been assigned a sequence number of 10. The second rule's sequence number is an increment of 10, and so on. To add a rule to the beginning of the existing ACL above, an administrator could issue the rule with a sequence-number value of 5. In addition, ACL keywords that map to specific port numbers have replaced those port numbers in the output. Port 23 has an equivalent keyword of telnet. The File Transfer Protocol (FTP) ports 20 and 21 have equivalent keywords of ftp-data and ftp, respectively.

The permit tcp any any range 22 443 command in this scenario creates a named extended ACL rule that permits a TCP packet from any source to any destination as long as that packet has a destination TCP port in the range from port number 22 through port number 443. The tcp protocol keyword configures the rule to match TCP packets. The source wildcard value of any configures the rule to match the TCP packet regardless of the source address. The destination wildcard value of any configures the rule to match the TCP packet regardless of the destination address. The range operator matches packets that have destination ports that are equal to the port number specified as the start of the range of ports, the port number specified as the end of the range of ports, and any port number that falls in between those two port numbers. The range operator is the only operator that requires two values.

Traffic that matches TCP destination port 23 will be permitted in this scenario because port number 23 falls within the range specified by the permit tcp any any range 22 443 command. The permit tcp any any range 22 443 rule in this scenario is immediately followed in the ACL by the deny tcp any any neq 23 rule, although traffic with a TCP destination port value of 23 need not match this rule to be permitted. The deny tcp any any neq 23 rule denies a TCP packet from any source to any destination if the packet has a destination TCP port that is not equal to port number 23. The neq operator matches traffic that is not equal to the port number that is specified after the operator, which is port 23 in this scenario. Any TCP packets that did not match the permit tcp any any range 22 443 rule in the ACL will match the deny tcp any any neq 23 rule because it covers the entire range of destination ports except for port 23. The neq operator matches traffic that is not equal to the port number that is specified after the operator, which is port 23 in this scenario.

Traffic that matches TCP destination port 20 or TCP destination port 21 will not be permitted in this scenario, because the permit tcp any any eq 20 21 rule follows the deny tcp any any neq 23 rule in the ACL. In addition, neither port 20 nor port 21 falls within the range specified by the permit tcp any any range 22 443 rule. If the deny tcp any any neq 23 rule were not configured, the permit tcp any any eq 20 21 rule would match TCP packets from any source to any destination as long as the TCP destination port is equal to either port 20 or port 21. The eq operator matches packets that have destination ports that are equal to the port number that is specified after the operator. In addition, up to 10 port numbers separated by spaces can be specified after the eq or neq operators.

Traffic that matches TCP destination port 444 will not be permitted in this scenario. Furthermore, packets with this TCP destination port values would not be permitted even if the deny tcp any any neq 23 rule were not configured. If the deny tcp any any neq 23 rule were not configured, the permit tcp any any gt 444 rule would match TCP packets from any source to any destination as long

as the TCP destination is greater than port number 444. The rule would not match packets that have a destination port of 444 or lower. The gt operator matches packets that have destination ports that are greater than the port number specified. Conversely, the lt operator matches packets that have destination ports that are less than the port number specified. Therefore, the permit tcp any any lt 442 command would not match packets that have a destination port of 442 or higher. In addition, the permit tcp any any lt 442 rule precedes the permit tcp any any gt 444 rule in the ACL. Only one port number value can be specified after the gt and lt operators. You cannot combine the gt and lt operators to specify a range.

Original Exam D Question 54

192.168. 1.2/28

192.168.1.1/28 192.168.1.3/28

a

RouterA

aa FT

RouterC

192.168.1.5/28 192.168.1.4/28

You administer the network shown above. You want to configure RouterA so that only RouterB can telnet to RouterA. No other traffic to or from RouterA should be affected.

Which of the following commands should you issue to complete this task? (Select 2 choices.)

RouterA(config-line)#access-class 10 in

RouterA(config-line)#access-class 10 out

RouterA(config-if)#ip access-group 10 out

RouterA(config-if)#ip access-group 10 in

RouterA(config)#access-list 10 permit 192.168.1.2 0.0.0.0
F.

RouterA(config)#access-list 10 permit 192.168.1.2 0.0.0.15

Correct Answer: A, E

Explanation

You should issue the following commands on RouterA:

RouterA(config)#access-list 10 permit 192.168.1.2 0.0.0.0
RouterA(config)#line vty 0 4

RouterA(config-line)#access-class 10 in

You can use access control lists (ACLs) to control Telnet traffic. The access-list 10 permit 192.168.1.2 0.0.0.0 command specifies that traffic from the host 192.168.1.2 should be permitted. The 0.0.0.0 in the access-list command is the wildcard mask. A wildcard mask is basically a reverse subnet mask that indicates which bits are ignored when determining whether an address is affected by the ACL. To calculate the subnet mask from a wildcard mask, or to calculate the wildcard mask from a subnet

mask, subtract the subnet mask or wildcard mask from 255.255.255.255. A wildcard mask of 0.0.0.0 indicates a specific host address.

If a packet does not match any of the statements in an access list, the packet is dropped. This is called the implicit deny rule; all traffic is dropped unless it is matched by one of the access list statements that is configured with the permit keyword. Therefore, no other hosts will be able to connect to RouterA by Telnet.

You should not issue the access-list 10 permit 192.168.1.2 0.0.0.15 command. A wildcard mask of 0.0.0.15 indicates that all hosts on the 192.168.1.0/28 network will match the ACL. Therefore, the access-list 10 permit 192.168.1.2 0.0.0.15 command would allow Telnet access from all of the routers on the network, because all of the routers' IP addresses fall within the 192.168.1.0/28 IP address range.

The access-class command applies an ACL to a virtual terminal (vty) line. The access-class 10 in command allows inbound Telnet access only from IP addresses in access list 10.

You should not issue the access-class 10 out command. The access-class 10 out command would allow outbound Telnet access from IP addresses in access list 10; it would not prevent inbound Telnet access.

You should not issue the ip access-group command. The ip access-group command is used to apply an ACL to an interface, not to a vty line. If you issued the ip access-group 10 in command in interface configuration mode, you would allow only traffic from access list 10 into the specified interface. If you issued the ip access-group 10 out command, you would allow only traffic from access list 10 out of the specified interface.

Day 36 - CDP & LLDP

Included Exam D items: Q34, Q59

Original Exam D Question 34

Which of the following statements about CDP and LLDP is not true? (Select the best answer.)

- A. They can each be disabled either globally or per interface.
- B. They each send advertisements to a multicast address.
- C. They are each a Cisco-proprietary discovery protocol.
- D. They each operate at the Data Link layer of the OSI model.

Correct Answer: C

Explanation

Cisco Discovery Protocol (CDP) is a Cisco-proprietary discovery protocol. Link-Layer Discovery Protocol (LLDP), on the other hand, is an Institute of Electrical and Electronics Engineers (IEEE) 802.1AB standard protocol. Both protocols advertise and discover devices on a local network. However, CDP discovers only Cisco devices. LLDP can be enabled on Cisco devices in order to discover network devices from other manufacturers.

Both CDP and LLDP can be disabled either globally or per interface. You should issue the no cdp run command from global configuration mode to disable CDP globally on a device. To re-enable CDP globally on a router or switch, you should issue the cdp run command. You should issue the no cdp enable command from interface configuration mode to disable CDP on a single interface. To re-enable CDP on an interface, you should issue the cdp enable command from interface configuration mode. Similar to CDP, LLDP can be disabled globally by issuing the no lldp run command. You can re-enable LLDP by issuing the lldp run command. On an interface, you can disable LLDP by issuing the no lldp enable command. You can re-enable LLDP on the interface by issuing the lldp enable command.

Both CDP and LLDP operate at the Data Link layer, or Layer 2, of the Open Systems Interconnection (OSI) network model. Because they operate at Layer 2, even devices using different Layer 3 protocols can communicate and share information. When troubleshooting CDP or LLDP, you should typically begin at the Data Link layer of the OSI model.

Both CDP and LLDP send advertisements to a multicast address. The CDP or LLDP advertisements contain Type, Length, and Value (TLV) fields that provide information about the sending device. The TLV fields and values that are sent in the advertisements depend on a number of factors, such as the device sending the advertisement and the version of CDP or LLDP that is running. For example, if LLDP is running without Media Endpoint Device (MED) extensions, Power over Ethernet (PoE) TLV fields like location and power management will not be sent.

Original Exam D Question 59

You need to discover the following information about a device connected to a Cisco switch:

- © The IP address of the neighboring device
- c The interface on the switch that is connected to the neighboring device

- © The interface on the neighboring device that is connected to the switch

Which of the following commands should you issue? (Select the best answer.)

- A. show cdp interface
- B. show cdp neighbors detail
- C. show cdp neighbors
- D. show cdp

Correct Answer: B

Explanation

You should issue the show edp neighbors detail command. Cisco Discovery Protocol (CDP) is an Open Systems Interconnection (OSI) Layer 2 Cisco-proprietary protocol that is used to advertise and discover only directly connected Cisco devices on a local network. By default, CDP sends advertisements every 60 seconds and retains information about a neighbor for 180 seconds before discarding it. Although CDP is enabled by default on many Cisco devices, it can be disabled for security purposes. However, disabling CDP may not be an option as some devices such as Cisco Voice over Internet Protocol (VoIP) phones rely on CDP to convey critical information, such as their power requirements, to other devices.

The show cdp neighbors detail command can provide detailed information about neighboring devices that have been discovered by using CDP. The show cdp neighbors detail command will provide the following information:

- © The device ID, or host name, of the neighboring device

- © The Internet Protocol (IP) address of the neighboring device
- © The platform, or product number, of the neighboring device
- © The capabilities of the neighboring device

- © The local interface

- © The remote interface

- © The holdtime

- © The software version

- © The native virtual local area network (VLAN)

c The VLAN Trunking Protocol (VTP) domain

The following is sample output from the show cdp neighbors detail command:

Switchi#show cdp neighbors detail

Device ID: boson-1
Entry address(es):
IP address: 10.1.2.3
Platform: WS-C2960-24TT-L, Capabilities: Switch IGMP
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/10
Holdtime : 117 sec

Version :

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 12.2(25)SEE2, RELEASE SOFTWARE (fc1)

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advertisement version: 2

VIP Management Domain: 'boson'
Native VLAN: 1

Duplex: full

Management address(es):

The show cdp neighbors command displays much of the same information found in the output of the show cdp neighbors detail command. However, it does not display the IP address of the neighboring device. The following information is displayed when the show cdp neighbors command is issued:

© The device ID of the neighboring device

© The capabilities of the neighboring device

e The product number of the neighboring device

© The holdtime

© The local interface

© The remote interface

The following displays sample output from the show cdp neighbors command:

Switchi#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater

Device ID Local Intrfce Holdtme Capability Platform Port ID

boson-1 Gig 0/1 117 SF WS-C2960-2 Gig 0/1

boson-2 Gig 0/2 143 # WS-C2960-4 Gig 0/1

boson-3 Fas 0/3 102 RSt 1841 Fas 0/1
boson-4 Fas 0/4 133 RSI 1841 Fas 0/2

The show cdp command displays global information about CDP, including timer and holdtime information. The following is sample output from the show cdp command:

```
Switch1#show cdp
```

Global CDP information:

Sending CDP packets every 60 seconds
Sending a holdtime value of 180 seconds

Sending CDPv2 advertisements is enabled

The show cdp interface command displays information about the interfaces on which CDP is enabled. The following is sample output from the show cdp interface command:

```
switch1#show cdp interface
```

GigabitEthernet0/0 is up, line protocol is up, encapsulation is ARPA
Sending CDP packets every 60 seconds
Holdtime is 180 seconds

FastEthernet0/0 is up, line protocol is up, encapsulation is ARPA
Sending CDP packets every 60 seconds
Holdtime is 180 seconds

Day 37 - NTP

Included Exam D items: Q51

Original Exam D Question 51

RouterA does not have an existing NTP configuration. You issue the ntp master command from global configuration mode on RouterA.

Which of the following statements is true regarding the NTP configuration on RouterA? (Select the best answer.)

- A. RouterA will use a default stratum value of 5.
- B. RouterA will respond to NTP queries on all of its active interfaces.
- C. RouterA will use a reference clock with an IP address of 127.0.0.1.
- D. RouterA will use NTP to update its hardware clock.

Correct Answer: B

Explanation

On a router without an existing Network Time Protocol (NTP) configuration, issuing the ntp master command from global configuration mode enables NTP on all active interfaces. The ntp master command configures a Cisco router to function as an NTP server and to use its software clock as an authoritative time source. In addition, if NTP has not been previously enabled, the ntp master command configures the router to respond to NTP queries on all active interfaces. The syntax of the ntp master command is ntp master [stratum], where stratum is an NTP stratum value from 1 through 15. By default, an NTP server is configured to use a stratum value of 8. The NTP stratum value is used by NTP clients to choose the most accurate time source. Devices with higher stratum numbers receive time from devices with lower stratum numbers. The lower the stratum number of a device, the closer the device is to an authoritative time source. For example, a stratum 2 device typically receives its time from a stratum 1 device, which is directly connected to an authoritative time source, such as an

atomic clock. A stratum 3 device typically receives its time from a stratum 2 device, and so on.

You can use the `show ntp associations` command to determine the reference clock that is used by an NTP device. When an NTP device uses its software clock as an authoritative time source, the Internet Protocol (IP) address of 127.127.7.1 is used for the value of the reference clock and IP address of the NTP server. For example, the following sample output indicates that RouterA is using its software clock as an authoritative time source:

```
RouterA#show ntp associations
```

```
address ref clock | st when poll reach delay offset disp
*~427.127.7.1 - 127.127.7.1 7 33 64 377 0.0 0.00 0.0
```

* master (syncd), # master (unsyncd), + selected, - candidate, ~ configured

The stratum value displayed for the reference clock will always be one stratum lower than the configured stratum value of the NTP server. In the sample output above, the reference clock is shown as having a stratum value of 7 because the `ntp master` command was issued without a stratum parameter and thus RouterA received the default stratum value of 8. You can issue the `show ntp status` command to verify the stratum value configured for a Cisco router. The following sample output indicates that RouterA is configured with the default stratum value of 8:

```
RouterA#show ntp status
```

```
Clock is synchronized, stratum 8, reference is 127.127.7.1
```

```
nominal freq is 249.5901 Hz, actual freq is 249.5901 Hz, precision is 2**18
reference time is BEB0044D.25D16B66 (16:27:57.147 EDT Fri May 18 2001)
clock offset is 0.0000 msec, root delay is 0.00 msec
```

```
root dispersion is 0.02 msec, peer dispersion is 0.02 msec
```

By default, a Cisco device configured to use NTP will synchronize its software clock with an NTP time source, but it will not synchronize its hardware clock. The software clock is used as the source of all time-related activities on a Cisco device, such as system log message timestamps; however, the time value is lost if the router is restarted. Most Cisco devices have a hardware clock in addition to a software clock. The hardware clock is used to initialize the software clock after a restart. Because the hardware clock is attached to an independent power source, such as a battery, it can preserve the date and time through restarts and periods of power loss. You can issue the `ntp update-calendar` command to configure a Cisco router to use NTP to update its hardware clock.

Day 38 - DNS

Included Exam D items: Q66

Original Exam D Question 66

Default_GW

Host. WWW_server

DNS_server

A web browser on HostA sends an HTTP request to WWW_server. This is the first time HostA has ever sent a request to WWW_server. HostA does not use a hosts file.

With which of the following devices does HostA establish a TCP connection in this scenario? (Select the best answer.)

- A. only WWW server
- B. DNS-server, Default_GW, and WWW_server
- C. only Default_GW and WWW_server
- D. only DNS_server and WWW_server

Correct Answer: A

Explanation

HostA establishes a Transmission Control Protocol (TCP) connection with only WWW_server in this scenario. If a host does not have the Internet Protocol (IP) address that corresponds to the fully qualified domain name (FQDN) of a web server in its cache and does not use a hosts file, it must first contact a Domain Name System (DNS) server to resolve the FQDN of the target web server into an IP address. The DNS server contains a hierarchical database of host name-to-IP address mappings.

Normal DNS requests and responses are sent by using User Datagram Protocol (UDP) and do not result in TCP connections with the DNS server. DNS can use TCP but typically generates TCP sessions for only zone transfers and query responses that exceed 512 bytes. This process is known as a recursive DNS search. Recursive DNS searches allow DNS servers to provide authoritative name resolution information to hosts.

Once the FQDN of the web server has been resolved into an IP address, the host will initiate the TCP three-way handshake to establish a TCP session with the web server. The host will send a zero-length SYN segment directed to the web server. The web server will respond with a zero-length SYN/ACK segment. To complete the handshake, the host will then respond with a zero-length ACK segment directed to the web server. Once the TCP session has been established, the host can finally send a Hypertext Transfer Protocol (HTTP) request initiated by the browser.

If the IP address is on the same subnet, the host will send its HTTP request directly to the web server at that IP address. If the IP address is on a different subnet, as in this scenario, the host will still send its HTTP request to the web server, but it will forward the IP packets to the host's default gateway instead of directly to the web server. Although the IP packets for the HTTP request are forwarded to the default gateway, this process is transparent to the TCP session over which the HTTP request is made. The default gateway does not modify the majority of the IP packet's header fields, such as the source and destination IP addresses, nor does it modify the payload of the IP packet, which includes the encapsulated TCP session. Instead, the gateway removes the Layer 2 header and trailer from the encapsulated IP packet and then adds a new Layer 2 header before forwarding the packet to the next hop on the route to the packet's final destination.

Day 39 - DHCP

Included Exam D items: Q32

Original Exam D Question 32

LAN B

You administer the network shown above. The following output is from the show ip interfaces brief command on RouterA:

```
RouterA#show ip  
Interface
```

```
FastEthernet0/0  
FastEthernete/1  
FastEthernet0/2  
FastEthernet0/3
```

interfaces brief

IP-Address

10.1.14.1

172.16.72.1

192.168.7.1

192.168.8.1

OK? Method

YES NVRAM

YES NVRAM

YES NVRAM

YES NVRAM

Status

up

up

up

up

Protocol

up

up

up

up

All addresses are configured with a 24-bit subnet mask.

Which of the following command sets should you use to configure RouterA to distribute IP address information to hosts on LAN B? (Select the best answer.)

RouterA(config)#ip dhcp pool LAN-B

A. RouterA(config-dhcp)#network 10.1.14.0 255.255.255.0

RouterA(config-dhcp)#default-router 10.1.14.1

RouterA(config)#ip dhcp pool porte-3

B. RouterA(config-dhcp)#network 192.168.8.0 255.255.255.0

RouterA(config-dhcp)#default-router 192.168.8.1

RouterA(config)#ip dhcp pool POOL3

C. RouterA(config-dhcp)#network 192.168.7.0 /24

RouterA(config-dhcp)#default-router 192.168.7.1

RouterA(config)#ip dhcp pool dhcp-B8

D. RouterA(config-dhcp)#network 172.16.72.0 /24

RouterA(config-dhcp)#default-router 172.16.72.1

Correct Answer: B

Explanation

You should issue the following command set to configure RouterA to use Dynamic Host Configuration Protocol (DHCP) to distribute Internet Protocol (IP) address information to hosts on LAN B:

RouterA(config)#ip dhcp pool porte-3

RouterA(config-dhcp)#network 192.168.8.0 255.255.255.0

RouterA(config-dhcp)#default-router 192.168.8.1

DHCP assigns IP address information to hosts based on the subnet to which the hosts belong. Therefore, you must determine what subnet is used on LAN B. LAN B is connected to the Fa0/3 interface of RouterA, which is configured with an IP address of 192.168.8.1. Consequently, the commands used to configure DHCP for hosts on LAN B must contain addressing information consistent with the 192.168.8.0/24 subnet.

The ip dhcp pool command establishes the DHCP pool. The syntax of the ip dhcp pool command is ip dhcp pool name, where name is the name of the DHCP pool. You can use any name for the DHCP pool. Therefore, the ip dhcp pool LAN-B, ip dhcp pool dhcp-B, ip dhcp pool POOL3, and ip dhcp pool port0-3 commands are all valid.

The network command indicates what IP address range should be assigned and what subnet should receive the addresses. The syntax of the network command is network address [mask | /prefix], where address is the network address, mask is the subnet mask, and prefix is the prefix length in Classless Inter-Domain Routing (CIDR) notation. Because you can use either the subnet mask or CIDR prefix, the network 192.168.8.0 255.255.255.0 and network 192.168.8.0 /24 commands are both valid.

The default-router command indicates the default gateway router address that should be configured on the hosts. The syntax of the default-router command is default-router address. Because the Fa0/3 interface of RouterA is connected to LAN B, the default gateway address for hosts on LAN B should be configured with the IP address of the Fa0/3 interface. Therefore, you should issue the default-router 192.168.8.1 command.

The following command set could be used to configure DHCP for hosts on LAN A:

```
RouterA(config)#ip dhcp pool LAN-B
RouterA(config-dhcp)#network 10.1.14.0 255.255.255.8
RouterA(config-dhcp)#default-router 10.1.14.1
```

The following command set could be used to configure DHCP for hosts on LAN C:

```
RouterA(config)#ip dhcp pool dhcp-B
RouterA(config-dhcp)#network 172.16.72.0 /24
RouterA(config-dhcp)#default-router 172.16.72.1
```

The following command set could be used to configure DHCP for hosts on LAN D:

```
RouterA(config)#ip dhcp pool POOL3
RouterA(config-dhcp)#network 192.168.7.0 /24
RouterA(config-dhcp)#default-router 192.168.7.1
```

Day 40 - SNMP

Included Exam D items: Q36

Original Exam D Question 36

Which of the following protocols is typically used to remotely monitor and manage network devices? (Select the best answer.)

- A. DHCP
- B. NAT

- C. SNMP
- D. NTP

Correct Answer: C

Explanation

Simple Network Management Protocol (SNMP) is used to remotely monitor and manage network devices. SNMP collects statistical data about devices, which administrators can use to monitor those devices. SNMP uses User Datagram Protocol (UDP) ports 161 and 162. SNMP version 3 (SNMPv3) is an enhanced version of SNMP that uses encryption to provide confidentiality, integrity, and authentication.

Network Time Protocol (NTP) is a management protocol that is used to ensure that syslog timestamps are accurate across multiple network devices. Using NTP to synchronize the time on network devices is useful for administering the devices. Because the time is synchronized, log files are displayed in the order in which they were created, which enables administrators to more accurately track when events occurred.

Network Address Translation (NAT) is used to translate Internet Protocol (IP) addresses on one network to IP addresses on another network. Typically, NAT is used to enable hosts on a privately addressed internal network to communicate with hosts on a public network, such as the Internet. Static NAT maps each internal translated IP address to a specified external IP address, whereas dynamic NAT allocates IP addresses from a pool as needed.

Dynamic Host Configuration Protocol (DHCP) is used to dynamically assign network configuration information to host devices. This network configuration information can include the IP address, subnet mask, default gateway, and Domain Name System (DNS) servers that the client computer will use.

Day 41 - Syslog

Included Exam D items: Q7

Original Exam D Question 7

You are attempting to troubleshoot interface protocol up/down transitions on a Cisco router. You want to display log messages to the console. However, you want to minimize the number of log messages as much as possible.

Which of the following commands should you issue? (Select the best answer.)

- A. logging console 6
- ☐ B. logging console 3
- C. logging console 4
- D. logging console 5

Correct Answer: D

Explanation

You should issue the logging console 5 command. You can filter log messages on the console by severity level by issuing the logging console severity-level command. Alternatively, you can filter log messages to a Syslog server by issuing the logging trap severity-level command. When the logging console or logging trap command is issued with a severity-level parameter, messages with the specified severity level and all lower-numbered severity levels will be displayed or sent, respectively.

Cisco debug messages and log messages are divided into the following severity levels:

© 0-emergencies

° 1-alerts

© 2-critical

e 3-errors

© 4-warnings

« 5 -notifications

© 6- informational

« 7-debugging

Messages are formatted in the Berkeley Software Distribution (BSD) Syslog format, which is a percent sign (%) followed by a facility code, a severity code, and a mnemonic code. The three codes are separated by dashes. Consider the following message:

"Dec 1 14:20:07.059: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

In this Syslog message, the facility code is LINEPROTO, the severity code is 5, which is equivalent to notifications, and the mnemonic code is upDoWN. The dash-separated code is followed by a colon and the human-readable text of the log message.

Route flaps, neighbor adjacencies, and interface protocol up/down transitions are displayed as notifications. These messages are at level 5; therefore, you must issue the logging console 5 command to display log messages at level 5 and below.

You should not issue the logging console 6 command. Level 6 is used to display messages at the informational level and below. Informational messages are routine messages but still might indicate a problem. Although line protocol up/down messages would be displayed by issuing the logging console 6 command, other less important messages would also be displayed.

You should not issue the logging console 4 command or the logging console 3 command. Level 4 is used to display messages at the warnings level and below; level 3 is used to display messages at the errors level and below. Critical, error, and warning messages indicate something that might impact the device. For example, interface up/down state changes are displayed as errors at level 3.

Emergencies, which are displayed at level 0, and alerts, which are displayed at level 1, indicate a severe hardware or software problem with the device. These messages need to be addressed immediately.

Day 42 - SSH & Device Management

Included Exam D items: Q57

Original Exam D Question 57

Drag the commands to the appropriate step in the process of enabling SSH on vty lines. Not all commands will be used.

Commands Steps

To enable Secure Shell (SSH) for virtual terminal (vty) lines on a Cisco router, you should complete the following steps:

1. Configure the router with a host name other than Router by issuing the hostname command.
2. Configure the router with a domain name by issuing the ip domain-name command.
3. Generate an RSA key pair for the router by issuing the crypto key generate rsa command.
4. Configure the vty lines to use SSH by issuing the transport input ssh command from line

configuration mode.

SSH is often used as a secure replacement for Telnet to manage network devices. In order for SSH to be enabled on a Cisco device, the device must be running a K9 IOS image, which provides cryptographic functionality.

It is not necessary to issue the `ip ssh version 2` command. By default, Cisco devices support both SSH version 1 (SSHv1) and SSH version 2 (SSHv2) connections. Issuing the `ip ssh version 2` command ensures that the device will accept only SSHv2 connections.

Reference:

Boson CCNA Curriculum 200-301, Module 7: IP Services, SSH Configuration

Cisco: Configure SSH on Routers: Set Up a Cisco IOS Router as SSH Client0

Category:

IP Services

**Correct Answer: 1. Configure a host name with the `hostname` command.
2. Configure a domain name with the `ip domain-name` command.
3. Generate an RSA key pair with `crypto key generate rsa`.
4. On the vty lines, allow SSH with `transport input ssh`.**

Explanation

To enable Secure Shell (SSH) for virtual terminal (vty) lines on a Cisco router, you should complete the following steps:

1. Configure the router with a host name other than Router by issuing the `hostname` command.
2. Configure the router with a domain name by issuing the `ip domain-name` command.
3. Generate an RSA key pair for the router by issuing the `crypto key generate rsa` command.
4. Configure the vty lines to use SSH by issuing the `transport input ssh` command from line configuration mode.

SSH is often used as a secure replacement for Telnet to manage network devices. In order for SSH to be enabled on a Cisco device, the device must be running a K9 IOS image, which provides cryptographic functionality.

It is not necessary to issue the `ip ssh version 2` command. By default, Cisco devices support both SSH version 1 (SSHv1) and SSH version 2 (SSHv2) connections. Issuing the `ip ssh version 2` command ensures that the device will accept only SSHv2 connections.

Days 44-45 - NAT

Included Exam D items: Q31

Original Exam D Question 31

You administer the network shown below:

LANB Serial 0/0

10.1.2.0/24

LANA

10.1.1.0/24

RouterA#show running-config

<output omitted>

interface FastEthernet0/0

ip address 10.1.2.1 255.255.255.0

ip nat inside

no fair-queue

!

interface FastEthernet0/1

ip address 10.1.1.1 255.255.255.0

ip nat inside

no fair-queue

!

interface Serial0/e

ip address 172.16.1.1 255.255.254.0

ip nat outside

!

router rip

network 10.0.0.0

network 192.168.200.0

!

ip nat pool nat-pool 172.16.1.2 172.16.1.21 prefix 23

ip nat inside source list 10 pool nat-pool

!

access-list 10 permit 10.1.0. 0.0.255.255

<output omitted>

Hosts connected to LAN A and LAN B access the external network through RouterA. Currently, only 20 users can simultaneously access the external network. Your supervisor has asked you to reconfigure RouterA so that all users from LAN A and LAN B can access the external network. You have also been instructed to use the minimum number of addresses from the 172.16.1.0/23 network while still providing access for all users.

Which of the following command sets will accomplish this task? (Select the best answer.)

RouterA(config)#no

A.

2 RouterA(config)#ip

© a, PouterA(config)tno

" RouterA(config)#ip

RouterA(config)#no

C. RouterA(config)#no

RouterA(config)#ip

ip nat inside source list 10 pool nat-pool

nat outside source list 10 pool nat-pool overload

ip nat inside source list 10 pool nat-pool

nat inside source list 10 pool nat-pool overload

ip nat pool nat-pool 172.16.1.2 172.16.1.21 prefix 23

ip nat inside source list 10 pool nat-pool

nat outside source list 10 interface serial 0/0 overload

RouterA(config)#no

D. RouterA(config)#no

RouterA(config)#ip

ip nat pool nat-pool 172.16.1.2 172.16.1.21 prefix 23

ip nat inside source list 10 pool nat-pool

nat inside source list 10 interface serial 0/0 overload

Correct Answer: D

Explanation

To configure Network Address Translation (NAT) on RouterA so that all users on LAN A and LAN B can access the external network, and to use the minimum number of Internet Protocol (IP) addresses from the 172.16.1.0/23 network, you should issue the following commands on RouterA:

```
RouterA(config)#no ip nat pool nat-pool 172.16.1.2 172.16.1.21 prefix 23
```

```
RouterA(config)#no ip nat inside source list 10 pool nat-pool
```

```
RouterA(config)#ip nat inside source list 10 interface serial 0/0 overload
```

NAT can map host addresses in one network to addresses in another network. In NAT terminology, one of these networks is designated as the inside network and the other is designated as the outside network, NAT is often used to allow hosts on a privately addressed internal network to communicate with hosts on the Internet. In this case, the private network is referred to as the inside network and the Internet is referred to as the outside network.

NAT can be configured to perform a one-to-one mapping, where an IP address from the outside network must be available for every host on the inside network. NAT can also be configured to allow a large number of hosts on the inside network to use a few IP addresses or even a single IP address from the outside network.

This scenario states that NAT is currently configured on the network, but only 20 hosts can simultaneously access the outside network; this configuration requires 20 public IP addresses from the outside network. In this case, the ip nat pool command is used to define a pool of addresses from the outside network to be used by hosts on the inside network when they need to communicate with the outside network. The ip nat inside source list command refers to an access list that defines which hosts on the inside network can use addresses from this pool when they are communicating with the outside network.

The scenario requires that you reconfigure NAT; all hosts on the inside network must be able to

communicate with the outside network and must use a minimum number of addresses from the outside network. This can be accomplished by using a technique called NAT overload. NAT overload allows multiple hosts on the inside network to communicate by using a single IP address, which is configured on the interface that connects to the outside network. A router configured with NAT maintains a port translation table that uses unused ports to map traffic from the single IP address on the outside network back to the hosts on the inside network. This translation table can be displayed when you issue the show ip nat translations command.

To configure NAT overload in this scenario, you should remove the NAT pool by issuing the no ip nat pool command, because the current NAT pool will no longer be used. You should also remove the ip nat inside source list command by issuing the no ip nat inside source list command. You should then issue the ip nat inside source list command, reference the interface that connects to the outside network, and use the overload keyword.

The ip nat inside source list 10 pool nat-pool overload command in the following block of commands does not minimize the number of IP addresses used from the 172.16.1.0/27 network unless the pool size is reduced:

```
RouterA(config)#no ip nat inside source list 10 pool nat-pool
```

```
RouterA(config)#ip nat inside source list 10 pool nat-pool overload
```

Although using the overload keyword is acceptable with a NAT pool, using the overload keyword with the pool in this scenario would consume 20 addresses, whereas using the overload keyword with an interface would consume only one address. Therefore, the overload keyword should be used with the ip nat inside source list 10 interface serial 0/0 command to force all traffic destined for the external network to be sourced with the IP address assigned to interface Serial 0/0 rather than sourced with IP addresses from a pool of addresses.

The ip nat outside command in the following block of commands is used to perform address translation on traffic that originates on the outside network and is destined for the inside network; this is the opposite of what you are trying to accomplish in this scenario:

```
RouterA(config)#no ip nat inside source list 10 pool nat-pool
RouterA(config)#ip nat outside source list 10 pool nat-pool overload
```

Likewise, the ip nat outside command in the following block of commands is used to perform address translation on traffic that originates on the outside network and is destined for the inside network:

```
RouterA(config)#no ip nat pool nat-pool 172.16.1.2 172.16.1.21 prefix 23
RouterA(config)#no ip nat inside source list 10 pool nat-pool
RouterA(config)#ip nat outside source list 10 interface serial 0/0 overload
```

Days 46-47 - QoS

Included Exam D items: Q55

Original Exam D Question 55

Which of the following is the first action performed on a packet when it is received on a QoS-enabled interface? (Select the best answer.)

- A. scheduling
- B. shaping
- C. queuing
- D. classification

Correct Answer: D

Explanation

Classification is the first action performed on a packet when it is received on a Quality of Service (QoS)-enabled interface. Network devices typically implement QoS measures based on various Data Link layer and Network layer traffic descriptors. When a packet enters the network, the first device that is capable of marking traffic examines the packet and places it into a traffic class. This process is called classification.

Once a packet has been placed into a traffic class, the packet is moved into the corresponding transmit buffer on the interface. If there is no congestion on the interface, the packet is transmitted immediately. However, when the interface receives more traffic than it can send, it can buffer the excess traffic in a software queue. Because each traffic class can be assigned its own queue, packets with different classifications are typically placed into separate queues. This process is called queuing.

Packets that have been queued are transmitted in accordance with the scheduling method configured on the interface. For example, if an interface has been configured to use custom queuing (CQ), each queue can send a fixed number of bytes before the next queue can send traffic. This process is called scheduling.

Traffic regulation techniques such as traffic shaping and traffic policing rely on the classification, queuing, and scheduling of packets in order to guarantee a level of service to certain classes of traffic. For example, traffic shaping buffers flows that exceed the configured thresholds. Because traffic is buffered to maintain a desired packet rate, erratic patterns are smoothed into a uniform flow and jitter is minimized.

Day 48 - Security Fundamentals

Included Exam D items: Q4, Q9, Q27, Q46, Q64, Q67, Q72, Q74

Original Exam D Question 4

You are responsible for configuring switches ASW and A512. To protect your switches from malicious users and accidental misconfiguration, you would like to achieve the following goals:

- + Configure the active access ports on ASW and ASW2 to immediately enter the err-disabled state after 30 seconds of inactivity.
- + Configure the inactive FastEthernet ports on ASW1 and ASW2 to be members of the VLAN named DEADPOOL, and ensure that these ports are not usable.
- + Create a local user on ASW1 and ASW2 with a username of Bob and a password of Bob12345.
- + Configure ASW and ASW2 for local AAA login authentication by default for authentication to the console and other serial interfaces. Ensure that each switch will use the user shared authentication servers.

Lab task - no separate correct-answer section in the source PDF.

Original Exam D Question 9

You issue the show running-config command on a router and receive the following partial output:

```
<output omitted>
aaa new-model
!
aaa authentication login default group tacacs+ none
aaa authentication login access group tacacs+ enable
aaa authorization exec default tacacs+
!
<output omitted>
```

```
!  
line con 0  
line aux 0  
line vty 0 4  
login authentication access  
transport input telnet
```

Which of the following statements is true regarding the authentication method that will be used when a user attempts to access the router by using a Telnet connection? (Select the best answer.)

- A. ARADIUS server on the network will be used to authenticate the user.
- B. The local user database will be used to authenticate the user.
- C. The user will only be required to enter the enable password to authenticate to the router.
- D. ATACACS+ server on the network will be used to authenticate the user.
- E. No authentication method will be used.

Correct Answer: D

Explanation

The Terminal Access Controller Access Control System Plus (TACACS+) server on the network will be used for authentication when a user attempts to access the router by using a Telnet connection. The `aaa authentication login` command specifies the authentication methods that are supported when a user logs in to the system. In this scenario, two `aaa authentication login` commands have been issued: `aaa authentication login default group tacacs+ none` and `aaa authentication login access group tacacs+ enable`. The `login authentication access` command has been issued for Telnet connections, so the `aaa authentication login access group tacacs+ enable` command will apply to Telnet connections. This command configures the router to first query a TACACS+ server to authenticate users. If the TACACS+ server were to become unavailable, the user would be required to enter the enable password to access the router. To prepare for the possibility that the authentication server might become unavailable, you should configure a router so that it will use another method, such as the local user database or the enable password, in the absence of the authentication server.

If the `none` keyword were listed first in the `aaa authentication login` command, no authentication method would be used; users would not be required to authenticate in order to access the router. Because the `login authentication access` command is applied to the virtual terminal (vty) line, Telnet connections will use the authentication method defined by the `aaa authentication login access group tacacs+ enable` command, which specifies that TACACS+ will be used for authentication. Thus users must be authenticated before accessing the router by using a Telnet connection.

A Remote Authentication Dial-In User Service (RADIUS) server will not be used in this scenario. Neither `aaa authentication login` command specifies that a RADIUS server be used for authentication.

The local user database will not be used in this scenario. Neither `aaa authentication login` command specifies that the local user database be used for authentication.

Because the `aaa authentication login access group tacacs+ enable` command is applied to Telnet connections, users will be required to enter the enable password to access the router if the TACACS+ server becomes unavailable. However, users will not have to authenticate by using the enable password unless the TACACS+ server becomes unavailable; nothing in this scenario indicates that the TACACS+ server is unavailable.

Original Exam D Question 27

Which of the following best describes authorization? (Select the best answer.)

- A. the process of verifying the level of access configured for a user
- B. the process of establishing a user's accounts upon hire
- C. the process of verifying a user's identity
- D. the process of recording the use of resources

Correct Answer: A

Explanation

Authorization is the process of verifying the level of access configured for a user. The following list defines the three phases of the Authentication, Authorization, and Accounting (AAA) process:

- © Authentication - the process of verifying a user's identity
- © Authorization - the process of verifying the level of access configured for a user
- © Accounting - the process of recording the use of resources

AAA systems manage user activity. AAA systems are typically more sophisticated than simple password authentication systems, such as a local password database. Two common AAA systems include Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access-Control System Plus (TACACS+).

RADIUS is a standard AAA protocol created by the Internet Engineering Task Force (IETF). Compared to TACACS+, RADIUS has several limitations. For example, RADIUS encrypts only the password in Access-Request packets; it does not encrypt the entire contents of the packet like TACACS+ does.

TACACS+ is a Cisco-proprietary protocol used during AAA operations. TACACS+ provides more granular and flexible control over user access privileges. For example, the AAA operations are separated by TACACS+, whereas RADIUS combines the authentication and authorization services into a single function. Because TACACS+ separates these functions, administrators have more control over access to configuration commands. In addition, TACACS+ encrypts the entire contents of packets, thus providing additional security.

Original Exam D Question 46

Which of the following Layer 2 attacks uses the MAC address of another known host on the network in order to bypass port security measures? (Select the best answer.)

- A. MAC flooding
- B. DHCP spoofing
- C. ARP poisoning
- D. MAC spoofing
- E. VLAN hopping

Correct Answer: D

Explanation

In a Media Access Control (MAC) spoofing attack, an attacker uses the MAC address of another known host on the network in order to bypass port security measures. MAC spoofing can also be used to impersonate another host on the network. Implementing port security with sticky secure MAC addresses can help mitigate MAC spoofing attacks.

In a MAC flooding attack, an attacker generates thousands of forged frames every minute with the intention of overwhelming the switch's MAC address table. A switch builds the MAC address table by analyzing the MAC addresses of ingress, or inbound, traffic on each port. However, once the MAC table is full, the switch can no longer make intelligent forwarding decisions and all traffic is flooded. This allows the attacker to view all data sent through the switch because all traffic will be sent out each port. Implementing port security can help mitigate MAC flooding attacks by limiting the number of MAC addresses that can be learned on each interface to a maximum of 128. A MAC flooding attack is also known as a Content Addressable Memory (CAM) table overflow attack.

In an Address Resolution Protocol (ARP) poisoning attack, which is also known as an ARP spoofing attack, the attacker sends a gratuitous ARP (GARP) message to a host. The GARP message associates the attacker's MAC address with the Internet Protocol (IP) address of a valid host on the network. Subsequently, traffic sent to the valid host address will go through the attacker's computer rather than directly to the intended recipient. Implementing Dynamic ARP Inspection (DAI) can help mitigate

ARP poisoning attacks.

In a virtual local area network (VLAN) hopping attack, an attacker attempts to inject packets into other VLANs by accessing the VLAN trunk and double-tagging 802.1Q frames. A successful VLAN hopping attack enables an attacker to send traffic to other VLANs without the use of a router. You can prevent VLAN hopping by disabling Dynamic Trunking Protocol (DTP) on trunk ports, by changing the native VLAN, and by configuring user-facing ports as access ports.

In a Dynamic Host Configuration Protocol (DHCP) spoofing attack, an attacker installs a rogue DHCP server on the network in an attempt to intercept DHCP requests. The rogue DHCP server can then respond to the DHCP requests with its own IP address as the default gateway address; hence all traffic is routed through the rogue DHCP server. You should enable DHCP snooping to help prevent DHCP spoofing attacks.

Original Exam D Question 64

Drag each security threat to the mitigation technique that is most likely to prevent it. Threats can be used only once. Use all threats.

Threats User Awareness or Physical Access
Training Control

**Correct Answer: User awareness or training -> brute-force attack; social engineering; pharming
Physical access control -> burglary; tailgating**

Explanation

Brute-force attacks, pharming, and social engineering can all be mitigated by implementing user awareness or training. Burglary and tailgating, on the other hand, are more likely to be mitigated by using physical access controls.

User awareness is a security program element that provides employees with information about protecting confidential data. User training is a security program element that familiarizes users with corporate security policies in a formal and mandatory fashion. Physical access control is a security program element that protects infrastructure locations such as network closets.

Educating users about enterprise security is an important step in ensuring the confidentiality and integrity of data. Through education, users without a technological background can learn to identify and thus avoid falling prey to common attacks. For example, users who have been educated on the dangers of phishing emails, pharming, and social engineering attacks will be less likely to click malicious links in emails than those who have not been educated on such tactics. Phishing is an attempt to compromise user information by disguising a malicious email as a legitimate email. Pharming is the use of a legitimate service to redirect users to a malicious or compromised site. Social engineering is the simple exploitation of human trust and instinctive behavior. The goal of both user awareness and user training is to minimize the likelihood of users becoming victims of their own instincts.

User awareness is different from user training in that user training is a more formal and mandatory process. Making users aware of potential threats might involve little more than sending emails or making announcements that describe the threat. User training, on the other hand, is formally instructing users by using a published company security policy that outlines how to identify a given threat and how to respond to it. An example of user training might be guiding users through a company's password complexity policy. In this case, users would be taught how to create secure passwords in order to mitigate common password attacks, such as a brute-force attack. A brute-force attack occurs when an attacker uses every possible combination of characters in an attempt to guess a password.

Although not specific to users, physical access control allows administrators to protect sensitive equipment and data from accidental or malicious compromise by users. For example, installing a

proximity badge reader on a network closet door prevents users, visitors, or malicious actors from accessing, corrupting, or stealing the hardware that keeps the local area network (LAN) running. Physical access controls can be as simple as a padlock or as elaborate as a mantrap, which is a system that is intended to prevent unauthorized people from following an authorized person into a secured location. A mantrap consists of a set of two locked doors that operate in a manner that prevents both doors from being opened at the same time. People who enter the mantrap must be identified before they are allowed to exit the mantrap into the secure area. Security guards can use facial recognition to verify the identity of proximity badge holders.

Original Exam D Question 67

An administrator has configured the clear-text password BO\$On! so that it must be used to access enable mode on a Cisco router. The password is encrypted in the running configuration. No other passwords configured on the device are encrypted.

Which of the following commands did the administrator most likely use to accomplish this goal? (Select the best answer.)

A. enable secret 5 BO\$On!

O 8. service password-encryption

C. enable secret BOSOn!

D. password 0 BO\$On!

Correct Answer: C

Explanation

Most likely, the administrator issued the enable secret BO\$On! command to configure the clear-text password BO\$On! so that it is used to access enable mode on the router and is encrypted in the running configuration. In some Cisco IOS versions prior to 15.3(3), the enable secret command by default stores an encrypted password in the device's configuration file by using a Secure Hash Algorithm (SHA) 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated.

because of a security flaw in th

The syntax for the enable secret command is enable secret [level level] {password | [encryption-type] encrypted-password}, where password is a string of characters that represents the clear-text password. Instead of supplying a clear-text password, you can specify an encryption-type value of 0, 4, or 5 and an encrypted-password value of either a clear-text password, a SHA-256 hash, or a Message Digest 5 (MD5) hash, respectively. Supplying a hash value requires that you have previously encrypted the value by using a hashing algorithm in the same fashion that IOS uses the algorithm. This command configures a password that is required in order to place the device into enable mode, which is also known as privileged EXEC mode. The device must, at a minimum, be placed into enable mode for the user to be able to display the running configuration.

The administrator did not issue the enable secret 5 BO\$On! command in this scenario. If the administrator had specified an encryption-type value of 5, the enable secret command would have assumed that the BO\$On! string was an MDS hash instead of a clear-text password. When issued, the command would have then generated an error message similar to the following:

ERROR: The secret you entered is not a valid encrypted secret.

To enter an UNENCRYPTED secret, do not specify type 5 encryption.

When you properly enter an UNENCRYPTED secret, it will be encrypted.

The administrator did not issue the service password-encryption command in this scenario. If the administrator had issued the service password-encryption command, all current and future passwords in the device's running configuration would be encrypted. In this scenario, only the password BO\$On! has been encrypted.

The administrator did not issue the password 0 BO\$On! command in this scenario. The line console 0 command followed by the password command configures a password for accessing the router by using the console. Typically, the console is accessed by physically connecting a console cable between the router and a device that is running terminal software. Issuing the password for the console places the device into user mode.

Original Exam D Question 72

Which of the following commands should you issue in order to encrypt the password that is required to access privilege level 7 on a router? (Select the best answer.)

- A. enable password 7 cisco
- B. enable password level 7 cisco
- C. enable secret level 7 cisco
- D. enable secret 7 cisco

Correct Answer: C

Explanation

You should issue the enable secret level 7 cisco command to encrypt the password that is required to access privilege level 7. In Cisco IOS 15, the enable secret command stores an encrypted password in the device's configuration file by using a Secure Hash Algorithm (SHA) 256-bit hash. The syntax for the enable secret command is enable secret [level level] {password | [encryption-type] encrypted-password}, where password is a string of characters that represents the clear-text password. Instead of supplying a clear-text password, you can specify an encryption-type value of 0, 4, or 5 and an encrypted-password value of either a clear-text password, a SHA-256 hash, or a Message Digest 5 (MD5) hash, respectively. Supplying a hash value requires that you have previously encrypted the value by using a hashing algorithm in the same fashion that IOS uses the algorithm. This command configures a password that is required in order to place the device into enable mode, which is also known as privileged EXEC mode. The device must, at a minimum, be placed into enable mode for the user to be able to display the running configuration. In some Cisco IOS versions prior to 15.3(3), the enable secret command by default stores an encrypted password in the device's configuration file by using a SHA 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated because of a security flaw in their implementation.

The optional level parameter specifies the privilege level for which the encrypted password should be used. You can create multiple privilege levels, from level 0 through level 15, to more granularly specify the commands that users can issue. In this scenario, you want to encrypt the password for privilege level 7. Consequently, you should issue the enable secret level 7 cisco command to configure cisco as the encrypted password to access privilege level 7.

You can issue the enable password command to create a password that must be used to gain access to privileged EXEC mode. The syntax of the enable password command is enable password

[level level] {password | [encryption-type] encrypted-password}. Thus issuing the enable password level 7 cisco command would configure cisco as the password required to access privilege level 7. However, unlike the enable secret command, the enable password command would configure the password to be stored as clear text.

If both the enable password command and the enable secret command are in the running configuration of a Cisco device, the device will ignore the password associated with the enable password command. If all passwords on the device have been encrypted by issuing the service password-encryption command, the router will still prefer the enable secret password over the enable password password.

Both the enable secret 7 cisco command and the enable password 7 cisco command would display an error because they contain invalid syntax. To create a password for a specific privilege level, you must include the level parameter.

Original Exam D Question 74

You issue the following command on RouterA:
RouterA(config)#enable secret BeSoNrtr

Which of the following is true regarding the configuration of RouterA? (Select the best answer.)

A. All existing and future enable, console, vty, and other passwords have been encrypted.

O 8B. The BOSoNrtr password has been applied to Telnet connections to RouterA.

RouterA has been configured to prevent unauthorized access to privileged EXEC

oc mode.

Remote access to RouterA has been limited to a single connection that requires the

D.

° password BOSONrtr.

Correct Answer: C

Explanation

RouterA has been configured to prevent unauthorized access to privileged EXEC mode. To prevent unauthorized access to the privileged EXEC mode, you should use the enable secret command, which stores the enable password as a Message Digest 5 (MD5) hash, or the less secure enable password command, which stores the password in clear text. By default, all Cisco passwords, except for the enable secret password, are stored as clear text unless the service password-encryption global configuration command is used. When the service password-encryption command is issued, existing passwords are encrypted and future passwords will be encrypted unless the no service password-encryption command is issued.

The password command is used to create a console, virtual terminal (vty), or auxiliary (AUX) line password. To configure a console password, enter line configuration mode by issuing the line console 0 command and then issue the password password command. To configure a vty password, enter line configuration mode by issuing the line vty start-line end-line command and then issue the password password command. To configure an AUX password, enter line configuration mode by issuing the line aux 0 command and then issue the password password command.

Telnet uses five vty lines, numbered 0 through 4, to log in to a router. Telnet is a terminal emulation protocol that runs over an Internet Protocol (IP) network. You should configure a password for Telnet connections to prevent unauthorized access. Telnet does not require a direct connection to the router and can be used to configure a router located anywhere on the network as long as network connectivity to the router being configured is available. The line vty 0 4 command places the router in line configuration mode and prepares vty lines 0 through 4 for configuration. To limit RouterA to a single Telnet connection, you would issue the line vty 0 command.

The login command enables password checking when a user connects to the router by using a vty connection, such as a Telnet connection. When the login command is issued along with the password command, users who attempt to access the router by using a vty connection will be prompted for a password. However, if the login command has not been issued, users will not be prompted for a password even if a password has been configured.

Day 49 - Port Security

Included Exam D items: Q58

Original Exam D Question 58

Which of the following statements is true regarding the action taken by an interface configured with the switchport port-security violation restrict command when more than the maximum number of addresses is learned? (Select the best answer.)

- A. It does not generate a syslog message.
- B. It does not place the interface into an err-disabled state.
- C. It does not generate an SNMP trap.
- D. It does not increment the violation counter.

Correct Answer: B

Explanation

The switchport port-security violation restrict command does not configure an interface to enter into an err-disabled state when more than the maximum number of addresses is learned. You can use port security to limit traffic on a switch port so that only traffic sent from one or more statically configured or dynamically learned source Media Access Control (MAC) addresses is allowed into a switch port. When port security is enabled on a port, a switch checks the source MAC address of inbound traffic to see whether that MAC address is authorized to send traffic through that switch port. The default action is to place the interface into an err-disabled state, which effectively shuts down the interface; therefore, interface FastEthernet 0/8 will shut down if more than one device attempts to send traffic through the port. This action will not affect any other ports on the switch.

You can issue the switchport port-security violation command to modify the action taken when more than the maximum number of addresses is learned. The switchport port-security violation command can be issued with either the restrict keyword, the protect keyword, or the shutdown keyword. The restrict keyword configures the interface to remain up if more than the maximum number of addresses is learned, but traffic from the violating devices is dropped and a log entry is generated. Similarly, the protect keyword configures the interface to remain up if more than the maximum number of addresses is learned and to drop traffic from violating devices; however, unlike the restrict keyword, the protect keyword does not create a log of the violations. The shutdown keyword configures the port to shut down if more than the maximum allowed MAC addresses are learned on the interface. The following table summarizes the actions an interface takes when configured to operate in one of the three violation modes:

Violation mode	Protect	Restrict	Shutdown
Forwards unauthorized traffic?	No	No	No
Generates SNMP trap?	No	Yes	No
Generates syslog message?	No	Yes	Yes
Increments violation counter?	No	Yes	Yes
Places port in err-disabled state?	No	No	Yes

Day 51 - Dynamic ARP Inspection

Included Exam D items: Q40

Original Exam D Question 40

You issue the show vlan brief command on Switch1 and receive the following partial output:

```
Switch1#show vlan brief
```

```
VLAN Name Status Ports
. default active Gie/1, Gie/2
11 ~- VLAN0011 active Fa0/1, Fa0/2, Fa0/3, Fae/4

Fa0/5, Fa0/6, Fa0/7, Fae/8
Fa0/9, Fae/10
```

12 VLAN0012 active Fa0/11, Fa0/12, Fa0/13, Fae/14
Fa0/15, Fa0/16, Fa0/17, Fae/18
Fae/19

14 ~VLAN0e814 active Fa0/20, Fa0/21, Fa0/22, Fae/23
Fae/24

<output omitted>

You issue the following commands on Switch1:

```
Switch1#configure terminal
```

```
Switch1(config)#ip arp inspection vlan 11-12,14  
Switch1(config)#interface range gigabitethernet 0/1 - 2  
Switch1(config-if-range)#switchport access vlan 11  
Switch1(config-if-range)#switchport mode access
```

```
Switch1(config-if-range)#ip arp inspection
```

Which of the following statements is true? (Select the best answer.)

- A. Only GigabitEthernet 0/1 and GigabitEthernet 0/2 are trusted ports.
- B. All ports on the switch are untrusted ports.
- C. Only VLAN 11 ports are trusted ports.
- D. All ports on the switch are trusted ports.
- E. Only GigabitEthernet 0/1 and GigabitEthernet 0/2 ports are untrusted ports.

Correct Answer: 8

Explanation

All the ports on Switch1 in this scenario are untrusted ports because the ip arp inspection trust command has not been issued anywhere in the scenario. By default, a port is configured as an untrusted port when Dynamic ARP Inspection (DAI) is enabled on that port. Therefore, configuring virtual local area networks (VLANs) 11, 12, and 14 with DAI by issuing the ip arp inspection vlan 11-12,14 command ensures that any port operating in those VLANs is automatically an untrusted port.

When DAI is configured on a switch, ports are labeled as either trusted or untrusted. An Address Resolution Protocol (ARP) packet that is received on a trusted port is not inspected, whereas an ARP packet received on an untrusted port is inspected. If an ARP reply received on an untrusted port does not match a trusted entry on the switch, the packet will be dropped.

When DAI is configured for an entire VLAN, you can override the default configuration for a given port by issuing the ip arp inspection trust command in interface configuration mode. In this scenario, the switchport access vlan 11 command configures both GigabitEthernet interfaces, when those interfaces are operating in access mode, to reside on VLAN 11. The switchport mode access command enables access mode on each interface. However, the ip arp inspection command, not the ip arp inspection trust command, has been issued on both the GigabitEthernet 0/1 interface and the GigabitEthernet 0/2 interface in this scenario. Therefore, the default configuration has not changed.

DAI can be enabled on a single VLAN or on multiple VLANs. To enable DAI, you should use the ip arp inspection vlan global configuration command. The syntax of the ip arp inspection vlan command is ip arp inspection vlan {vian-/D | vian-range}. A range of VLANs can be entered by using a comma-separated list and/or a dash-separated pair of VLAN numbers indicating the range of VLANs to include. For example, each of the following commands enables DAI on VLANs 2 through 4: ip arp inspection vlan 2,3,4 and ip arp inspection vlan 2-4. Configuring DAI on each VLAN ensures that traffic sent from each host is inspected. In addition, each port is by default an untrusted port.

Day 53 - WAN Architectures

Included Exam D items: Q35, Q50

Original Exam D Question 35

Which of the following is true about IPSec configured in tunnel mode? (Select the best answer.)

- A. It encrypts the entire packet.
- B. It does not require an additional header.
- C. It does not encrypt the IP header.
- D. It is required for NAT traversal.

Correct Answer: A

Explanation

Internet Protocol Security (IPSec) configured in tunnel mode encrypts the entire packet. IPSec is a suite of protocols that can be used to encrypt Generic Routing Encapsulation (GRE) tunnel traffic, such as over a virtual private network (VPN). IPSec supports two modes: transport and tunnel. In tunnel mode, IPSec encrypts the entire Internet Protocol (IP) packet, including the header.

IPSec configured in tunnel mode does require an additional header. This is because the entire packet is encrypted. Layer 3 forwarding devices are not capable of decrypting the encrypted packet. Therefore, a new unencrypted IP header encapsulates the encrypted packet for routing.

IPSec configured in transport mode, not tunnel mode, does not encrypt the IP header. In transport mode, only the IP packet's payload is encrypted by IPSec, which means that the IP packet's header remains intact.

IPSec tunnel mode is not required for Network Address Translation (NAT) traversal. However, IPSec tunnel mode is more compatible with NAT than IPSec transport mode. IPSec in tunnel mode encrypts the entire packet, including both the header and payload. The encrypted packet is then encapsulated in a new IP packet that includes a new IP header, which can be used by NAT. Transport mode, on the other hand, creates complications for NAT if Authentication Header (AH) is used. It is possible to use IPSec in transport mode when NAT is deployed. However, it requires the use of a NAT Traversal (NAT-T) solution as described in the Internet Engineering Task Force (IETF) Request for Comments (RFC) 3947.

Original Exam D Question 50

What is the total bandwidth of a T3 connection? (Select the best answer.)

- A. 44.736 Mbps
- B. 2.048 Mbps
- C. 34368 Mbps
- D. 1.544 Mbps

Correct Answer: A

Explanation

AT3 connection has a total bandwidth of 44.736 megabits per second (Mbps). Primarily used in the United States and Japan, T3 connections transmit data across 672 DSO channels 8,000 times per second. T3 connections are usually too cost prohibitive for use except by Internet and telephone service providers.

AT1 connection has a total bandwidth of 1.544 Mbps. To achieve the 1.544-Mbps rate, T1 connections encapsulate data in DS1 signaling frames. DS1 signaling frames include 24 DSO channels and 1 framing bit. Each of the 24 DSO channels transfers 8 bits of data at a time. The inclusion of the framing bit raises the total amount of data in a single DS1 frame to 193 bits. T1 connections transmit 8,000 DS1 frames per second, which brings the total bandwidth to 1,544,000 bits per second (bps), or

1.544 Mbps. T1 connections typically transfer data across two pairs of shielded copper wires, known as a T1 line, which extend from the service provider to the customer. One pair of wires within the T1 line is used to send data, whereas the other pair is used to receive data. T1 is primarily used in the United States and Japan.

An E1 connection has a total bandwidth of 2.048 Mbps. Primarily used in Europe, E1 connections operate similarly to T1 connections. However, instead of using 24 eight-bit DSO channels, E1 connections use 32 eight-bit timeslots. E1 connections transmit 8,000 timeslot segments per second, which equals 2,048,000 bps, or 2.048 Mbps. Like T1s, E1 connections transfer data across two pairs of shielded copper wires.

An E3 connection has a total bandwidth of 34.368 Mbps. Primarily used in Europe, E3 connections consist of 16 E1 lines. E3 connections are usually too cost prohibitive for use except by Internet and telephone service providers.

Day 55 - Wireless Fundamentals

Included Exam D items: Q21, Q39

Original Exam D Question 21

Which of the following definitions best describes amplitude? (Select the best answer.)

- A. the length of a wave from crest to crest or trough to trough
- B. the number of waves per time period
- C. the height of a wave from trough to crest
- D. the length of a wave from crest to trough

Correct Answer: C

Explanation

Amplitude is best described as the height of a wave from trough to crest. In voice technology, amplitude is the change in voltage of an analog voice waveform during one wave cycle. A wave cycle is the progress of a wave from zero voltage, through its crest and trough, and back to zero. Amplitude can be measured as peak amplitude or as peak-to-peak amplitude. Peak amplitude is measured from zero voltage to the crest of a wave or from zero voltage to the trough of a

wave. Peak-to-peak amplitude is measured from the height of the wave cycle at the crest to the depth of the trough at the bottom, as illustrated by the following exhibit:

Voltage

Time

Wavelength, not amplitude, is the length of a wave from crest to crest or trough to trough. The wavelength can be measured between wave crests, between wave troughs, or between corresponding Zero crossings; Zero crossings are the points where the voltage changes from negative to positive or positive to negative. Wavelength is illustrated by the following exhibit:

Time

Amplitude is not the length of a wave from crest to trough. Because wavelength is measured from crest to crest or trough to trough, the length of a wave from crest to trough is half a wavelength.

Frequency, not amplitude, is the number of waves per time period. Electromagnetic waves are typically measured in cycles per second (cps). The international standard for cps is hertz (Hz), which is defined as 1 cps. For example, if the time period displayed in the exhibits above is equal to one

second, the frequency could be measured as 1.5 wave cycles per second or 1.5 Hz. The frequency is inversely proportional to the wavelength; as the wavelength decreases, the frequency increases.

Original Exam D Question 39

You are configuring a new wireless AP for a small office in Dallas. The AP uses the 2.4-GHz frequency band. The wireless APs in neighboring offices are using channels 1 and 6.

Which of the following channels should you configure the new wireless AP to use? (Select the best answer.)

OA 4

Correct Answer: D

Explanation

Of the choices provided, you should configure the new wireless access point (AP) to use channel 11. A channel is a small portion of the spread-spectrum used for transmission. APs operating on the same channel and within close physical proximity to other APs may experience some interference. To avoid potential interference while continuing to offer the same physical range of coverage, technicians can configure the channel used by the individual APs to a channel farther away on the spread-spectrum than the channel being used by the other AP in the same service radius.

By default, APs that use the 2.4-gigahertz (GHz) spectrum, which include those on 802.11b, 802.11g, 802.11n, 802.11ax, and 802.11be networks, typically use the nonoverlapping channels 1, 6, or 11 to decrease the potential for interference issues by keeping the channels in use evenly spread apart. The following table displays the channels that can be used within the 2.4-GHz band in the United States and Canada, along with their respective frequency ranges:

Channel	Range (GHz)
2	2.406 - 2.428
3	2.411 - 2.433
4	2.416 - 2.438
5	2.421 - 2.443
7	2.431 - 2.453
8	2.436 - 2.458
9	2.441 - 2.463
10	2.446 - 2.468

You should not configure the new wireless AP to use channels 3 or 4, because these channels interfere with channels 1 and 6. You should not configure the new wireless AP to use channel 9, because channel 9 interferes with channel 6. You cannot configure the new wireless AP to use channel 13 in North America; however, channels 1 through 13 can be used in Europe and Japan.

Day 56 - Wireless Architectures

Included Exam D items: Q16, Q17

Original Exam D Question 16

Which of the following statements are true regarding a lightweight AP that is operating in FlexConnect mode? (Select 2 choices.)

A. The lightweight AP can be remotely managed over a WAN link.

Os.

The lightweight AP must be connected to a WLC in order to operate.

The lightweight AP is running in the default lightweight AP mode.

The lightweight AP can be used to detect rogue clients on a wired network.

The lightweight AP can support client connections.

Correct Answer: A, E

Explanation

A lightweight access point (AP) that is operating in FlexConnect mode can support client connections and can be remotely managed over a wide area network (WAN) link. After adding a lightweight AP to a wireless LAN controller (WLC), you can configure the mode of the lightweight AP depending on your needs and the capability of the lightweight AP. A lightweight AP operating in FlexConnect mode can connect over a WAN link to a WLC that is located in a different location, such as a remote branch office. This enables administrators to manage the lightweight AP from a central location without having to deploy WLCs to each remote office. Furthermore, lightweight APs operating in FlexConnect mode can provide client connectivity even if the connection to the remote WLC is lost. That is, a lightweight AP operating in FlexConnect mode can authenticate clients locally even if the lightweight AP cannot reach the WLC.

A lightweight AP operating in FlexConnect mode is not running in the default lightweight AP mode. A lightweight AP is in local mode by default. In this mode, the lightweight AP can connect to a WLC and can provide client connectivity. In addition, a lightweight AP operating in local mode monitors all wireless channels.

A lightweight AP operating in FlexConnect mode is not used to detect rogue clients on a wired network. You can configure a lightweight AP to operate in rogue detector mode to configure the lightweight AP to scan traffic on the wired connection in search of rogue lightweight APs and rogue clients on the wired network.

Original Exam D Question 17

Which of the following statements is not true regarding a WLC service port? (Select the best answer.)

- A. An IPv4 address can be statically configured.
- B. An IPv6 address can be dynamically configured.

C. The service port IP address cannot be in the same subnet as the management interface IP address.

- D. A default gateway can be assigned through DHCP.

Correct Answer: D

Explanation

A default gateway cannot be assigned to the service port interface of a wireless LAN controller (WLC) through Dynamic Host Configuration Protocol (DHCP). A WLC can contain up to four static interfaces: the management interface, the AP-manager interface, the virtual interface, and the service port interface. The service port interface is used for out-of-band maintenance purposes on a WLC and can be accessed by using Telnet or Secure Shell (SSH). This interface is a physical interface on the WLC that can be used to recover the WLC in the event that the WLC fails.

The service port can be configured with a static Internet Protocol version 4 (IPv4) or IP version 6 (IPv6) address or it can be assigned a dynamic IP address through DHCP for IPv4 addresses or Stateless Address Auto-Configuration (SLAAC) for IPv6 addresses. Although the service port can automatically obtain an IP address, it cannot be assigned a default gateway. Instead, static routes can be manually configured on the WLC if the service port must be accessed from remote subnets. In addition, the service port IP address cannot be in the same subnet as IP address of the management interface.

The virtual interface supports mobility management by providing a specific IP address that is the

same across multiple controllers when wireless clients roam among the controllers. This enables seamless roaming among the controllers. The virtual interface supports embedded Layer 3 security. For example, the virtual interface can be used in situations where web authorization has been enabled for clients; the user is redirected to the IP address of the virtual interface when the user opens a web browser. In addition, if Dynamic Host Configuration Protocol (DHCP) relay has been enabled on the controller, the virtual interface can be used as the DHCP server address on wireless clients.

The AP-manager interface contains the IP address that is used as the source IP address by which the lightweight access points (APs) communicate with the WLC. Because the AP-manager interface communicates with the lightweight APs on the wireless network, the IP address assigned to the AP-manager interface should be unique on the network. After the interface has been configured, the WLC uses the AP-manager interface to listen for Layer 3 Lightweight Access Point Protocol (LWAPP) communications.

The management interface is used for in-band management information. This interface is used for all Layer 2 LWAPP communications between the controller and the lightweight APs. In addition, the management interface is used to communicate with other WLCs on the wireless network.

In addition to the four static interfaces, a WLC can contain up to 512 dynamic interfaces. Dynamic interfaces are user-defined and are typically used for wireless client data. The dynamic interfaces function similarly to virtual local area networks (VLANs). For example, you can create a dynamic interface to segment traffic on the WLC.

Day 57 - Wireless Security

Included Exam D items: Q12, Q61

Original Exam D Question 12

You want to increase the security of your WLAN. You decide to disable SSID broadcasts, enable MAC filtering, and enable a security method that supports AES-GCMP.

Which of the following configuration changes will be the most effective in increasing security? (Select the best answer)

- A. enabling WEP
- B. enabling WPA3
- C. enabling WPA2
- D. enabling WPA

Correct Answer: B

Explanation

Of the choices provided, enabling Wi-Fi Protected Access 3 (WPA3) will be the most effective in increasing the security of your wireless local area network (WLAN). WPA3 was introduced by the Wi-Fi Alliance in 2018 to address weaknesses in WPA2. WPA3 improves encryption by introducing support for Advanced Encryption Standard-Galois/Counter Mode Protocol (AES-GCMP). However, WPA3-Personal mode still supports using Advanced Encryption Standard-Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (AES-CCMP) at a minimum. The WPA3 standard introduced support for Protected Management Frames (PMF), which secures management frame communication between an access point (AP) and a client. For example, with PMF a wireless LAN controller (WLC) can be configured to use an Association Comeback Timer to mitigate Denial of Service (DoS) attacks. In addition, the WPA3 standard introduces Simultaneous Authentication of Equals (SAE), which enables clients and APs to mutually authenticate. Like Wi-Fi Protected Access (WPA) and Wi-Fi Protected Access 2 (WPA2), WPA3 also supports 802.1X and EAP-based authentication methods.

Enabling WPA2 will not be the most effective in increasing the security of your WLAN in this scenario,

because WPA2 does not support AES-GCMP. WPA2 consists of an authentication mechanism, an encryption mechanism, and an integrity mechanism. The authentication methods supported by WPA2 are 802.1X authentication, which is also known as WPA2 Enterprise, and pre-shared keys (PSKs), which is also known as WPA2 Personal. The encryption method supported by WPA2 is Advanced Encryption Standard (AES), which provides 256-bit encryption. The integrity mechanism supported by WPA2 is Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP).

Enabling WPA will not be the most effective in increasing the security of your WLAN in this scenario, because WPA does not support AES-GCMP. The Wi-Fi Alliance developed WPA to address significant security weaknesses in the Wired Equivalent Privacy (WEP) protocol. Although WPA resolved many of the weaknesses associated with WEP, it was only intended as an interim solution until the Institute of Electrical and Electronics Engineers (IEEE) 802.11i standard, which is implemented by WPA2, was certified and implemented. WPA uses Temporal Key Integrity Protocol (TKIP) for encryption, which is a stronger encryption algorithm than the method that WEP uses. TKIP uses the RC4 algorithm as its cipher. TKIP provides security features such as key hashing, a message integrity check, and a long initialization vector (IV). Keys used by TKIP can change dynamically, which increases wireless network security. TKIP provides 128-bit encryption. In addition, WPA supports 802.1X and Extensible Authentication Protocol (EAP)-based authentication methods. EAP is an authentication framework that can be used for point-to-point connections and for wired and wireless links.

Enabling WEP will not be very effective in increasing the security of your WLAN. Although implementing WEP is better than implementing no security at all, a WEP key can be easily cracked because it has a short, 24-bit IV. The IV value is added to a 40-bit WEP key to provide 64-bit encryption and is added to a 104-bit WEP key to provide 128-bit encryption. Because the IV value is sent as clear text, it can be intercepted and used to decrypt packets or it can be used for IV replay attacks. WEP is also subject to bit-flipping attacks, which exploit a weakness in the Integrity Check Value (ICV). WEP supports only static, pre-shared keys.

Original Exam D Question 61

Which wireless security protocol relies on SAE and introduces support for PMF? (Select the best answer.)

- A. WPA3-Personal
- ☐ B. WPA2-Enterprise
- C. WPA2-Personal
- D. WPA3-Enterprise

Correct Answer: A

Explanation

Wi-Fi Protected Access 3 (WPA3)-Personal is the wireless security protocol that relies on Simultaneous Authentication of Equals (SAE) and introduces support for Protected Management Frames (PMF). SAE in WPA3-Personal replaces the use of pre-shared keys (PSKs) in WPA2-Personal. WPA3 was introduced by the Wi-Fi Alliance in 2018 to address weaknesses in WPA2. WPA3-Personal mode supports using Advanced Encryption Standard (AES)-Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) at a minimum. PMF secures management frame communication between an access point (AP) and a client. SAE enables clients and APs to mutually authenticate.

WPA3-Enterprise improves authentication encryption by introducing support for 256-bit AES-Galois/Counter Mode Protocol (GCMP). Key establishment and authentication is accomplished by using Elliptic Curve Diffie-Hellman (ECDH) Exchange and Elliptic Curve Digital Signature Algorithm (ECDSA) with a 384-bit curve.

WPA2 was created by the Wi-Fi Alliance as an improvement to the original WPA standard and fully implements the Institute of Electrical and Electronics Engineers (IEEE) 802.11i standard. WPA2 uses AES-CCMP for data encryption. AES-CCMP provides much stronger encryption than RC4, which is the underlying encryption protocol used by both WPA and WEP; AES-CCMP supports an encryption key

length of up to 256 bits. WPA2-Enterprise, which uses 802.1X authentication, includes snooping protections and enhanced security methods that are not available in WPA2-Personal, which relies on PSKs.

Day 58 - Wireless Configuration

Included Exam D items: Q63, Q73

Original Exam D Question 63

You are configuring security on a new WLAN by using the WLC GUI.

Which of the following security settings are you most likely to configure by using the Layer 3 Security drop-down list box on the Layer 3 tab? (Select the best answer)

A. VPN Pass-Through

O 8. Web Authentication

C. Web Passthrough

D. WPA+WPA2

Correct Answer: A

Explanation

When you are configuring a new wireless local area network (WLAN), you are most likely to configure the VPN Pass-Through setting by using the Layer 3 Security drop-down list box on the Layer 3 tab of the Cisco wireless LAN controller (WLC) graphical user interface (GUI). There are two types of WLANs that you can configure by using the WLC GUI: a WLAN and a Guest LAN. The VPN Pass-Through setting is only available when you are configuring a WLAN.

When you configure a new WLAN by using the WLC GUI, you can configure security settings by clicking the new WLAN's Security tab. By default, the Layer 2 tab is selected when you click the Security tab. However, it is not possible to configure Layer 2 security on a Guest LAN.

On the Layer 2 tab of the Security tab, you can select one of the following Layer 2 wireless security features from the Layer 2 Security drop-down list box:

© None, which disables Layer 2 security and allows open authentication to the WLAN

© WPA+WPAZ, which enables Layer 2 security by using Wi-Fi Protected Access (WPA) or the more secure WPA2

© 802.1X, which enables Layer 2 security by using Extensible Authentication Protocol (EAP) authentication combined with a dynamic Wired Equivalent Privacy (WEP) key

© Static WEP, which enables Layer 2 security by using a static shared WEP key

© Static WEP + 802.1X, which enables Layer 2 security by using either a static shared WEP key or EAP authentication

© CKIP, which enables Layer 2 security by using the Cisco Key Integrity Protocol (CKIP)

« None + EAP Passthrough, which enables Layer 2 security by using open authentication combined with remote EAP authentication

There are two different sets of Layer 3 security features that you can configure on a Cisco WLC: one set for a WLAN and one set for a Guest LAN. Depending on which type of WLAN you create and which Layer 2 security options you have selected, you can select one of the following Layer 3 wireless

security features from the Layer 3 Security drop-down list box on the Layer 3 tab of the Security tab in the WLC GUI:

© None, which disables Layer 3 security no matter which Layer 2 security option is configured and regardless of whether you are configuring a WLAN or a Guest LAN

© IPSec, which enables Layer 3 security for WLANs by using Internet Protocol Security (IPSec)

« VPN Pass-Through, which enables Layer 3 security for WLANs by allowing a client to establish a connection with a specific virtual private network (VPN) server

© Web Authentication, which enables Layer 3 security for Guest LANs by prompting for a user name and password when a client connects to the network

* Web Passthrough, which enables direct access to the network for Guest LANs without prompting for a user name and password

Not every Layer 3 security mechanism is compatible with every Layer 2 security mechanism. It is therefore important to first configure Layer 2 security options before you attempt to configure Layer 3 security options.

Original Exam D Question 73

You are creating a new WLAN profile. You want only 5-GHz clients to be able to connect and transmit data at up to 54 Mbps. You use the GUI to navigate to the General tab of the WLANs page.

Which of the following parameters should you set for the Radio Policy option? (Select the best answer.)

- A. 802.11a/g only
- B. 802.11a only
- C. 802.11g only
- D. 802.11b/g only

Correct Answer: B

Explanation

You should set the 802.11a only parameter for the Radio Policy option. To ensure that the newly created WLAN profile is active, you must also select the checkbox next to Enabled for the Status option.

The Radio Policy option, which is found on the General tab of the WLANs page, is used to allow only clients that use certain radio spectra to communicate on a wireless local area network (WLAN). The 802.11a wireless standard uses the 5-gigahertz (GHz) operating frequency range and transmits data at up to 54 megabits per second (Mbps).

You should not set the 802.11g only parameter for the Radio Policy option. The 802.11g wireless standard uses the 2.4-GHz operating frequency range and transmits data at up to 54 Mbps.

You should not set the 802.11a/g only parameter for the Radio Policy option. Although the 802.11g standard allows for the transmission of data at 54 Mbps, it does not use the 5-GHz spectrum.

You should not set the 802.11b/g only parameter for the Radio Policy option. The 802.11b wireless standard uses the 2.4-GHz operating frequency range and transmits data at up to 11 Mbps. Therefore, neither 802.11b nor 802.11g use the 5-GHz spectrum.

Day 60 - JSON, XML & YAML

Included Exam D items: Q24, Q29

Original Exam D Question 24

A REST API query returns the following output:

```
"id": 12345,  
John",  
"lname": "Doe",  
"group": {  
  "role"  
  
  "fname"  
  
  "Receivables",  
  
  "read-only ": [  
    "Accounting Folder ",  
    "sales Folder "
```

Which of the following statements are true? (Select 2 choices.)

- A. The value of the group key is an object.
- B. The value of the read-only key is a text value.
- C. The value of the role key is an array.
- D. The value of the id key is numeric.

Correct Answer: A, D

Explanation

Of the available choices, it is true that the value of the group key in the Representational State Transfer (REST) Application Programming Interface (API) output above is an object. In addition, the value of the id key is numeric. REST is an API architecture that uses Hypertext Transfer Protocol (HTTP) or HTTP Secure (HTTPS) to enable external resources to access and make use of programmatic methods that are exposed by the API. The REST API that is being queried in this scenario has returned output in JavaScript Object Notation (JSON) format. JSON is a data modeling language that is commonly used by REST APIs. The JSON data modeling language returns data in the form of an object that contains key and value pairs.

The value of the id key is numeric in the JSON object in this scenario. Numeric data is defined by digits that are not enclosed in quotation marks. If a string of digits is enclosed in quotation marks, it will be treated as text even if it contains only numeric characters. The numeric value of the id key in this scenario is 12345.

A single JSON object can contain multiple key and value pairs. Each key and value pair inside a JSON object is separated from the others by a comma (.). Furthermore, each pair's key is separated from its value by a colon (:). The element in quotation marks on the left side of each colon is the key. The element on the right side of each colon is the value, which might or might not be enclosed in quotation marks. There are several data value types that can be returned in JSON output: text, numeric, array, object, Boolean, and null.

The value of the lname key, the value of the fname key, and the value of the role key are all text values in the JSON object in this scenario. JSON keys and JSON text values are always enclosed in quotation marks. Text values typically contain alphanumeric or Unicode characters.

The value of the group key in this scenario is another JSON object. A JSON object can be identified by the curly brackets that mark the beginning and the end of the object. A left curly bracket ({) marks the beginning of a JSON object. A right curly bracket (}) marks the end of a JSON object. JSON objects can be nested as values inside other JSON objects. In addition, JSON objects can be specified as values inside other JSON objects. In this scenario, the value of the group key in the primary JSON object is shown in the following JSON object:

```
{
  "role": "Receivables",
  "read-only": [
    "Accounting Folder",
    "Sales Folder"
  ]
}
```

As previously mentioned, a JSON object is a group of key and value pairs. The JSON object above contains two key and value pairs. The role key contains a text value of Receivables. The read-only key, on the other hand, contains an array.

The value of the read-only key in this scenario is an array that contains two text values: Accounting Folder and Sales Folder. Unlike JSON objects, arrays contain values only. They do not contain key and value pairs. Arrays are typically contained within square brackets. A left square bracket (D) indicates the beginning of the array. A right square bracket (J) indicates the end of the array. Each value inside an array is separated from the others by a comma (.). Although the array in this scenario contains two text values enclosed in quotation marks, an array can contain any of the other JSON data types. For example, an array can include JSON objects or other arrays as values. In addition, a given array need not contain only a single data type.

There are no Boolean or null values in the JSON output in this scenario. A given JSON key can be assigned a Boolean value of either true or false. Because these are Boolean values, not text values, they are typically not placed inside quotation marks. Null values technically have no value at all, although it is not exactly the same as a numeric value of 0. A given JSON key can be assigned a null value by specifying null without quotation marks.

Original Exam D Question 29

Which of the following is a REST API encoding format that uses HTML-like tags to define blocks of data? (Select the best answer.)

- A. XML
- B. YAML

- C. BSON
- D. JSON

Correct Answer: A

Explanation

Of the available choices, Extensible Markup Language (XML) is a Representational State Transfer (REST) Application Programming Interface (API) encoding format that uses Hypertext Markup Language (HTML)-like tags to define blocks of data. REST APIs encode data in either XML format or in JavaScript Object Notation (JSON) format. In addition, REST APIs are typically used to communicate with a Software-Defined Networking (SDN) application plane.

An SDN controller uses two different sets of APIs: one set to communicate with applications and another set to communicate with devices. Northbound APIs enable an SDN controller to communicate with applications in the application plane. Applications use northbound APIs to send requests or instructions to the SDN controller, which uses that information to modify and manage network flow. Southbound APIs enable an SDN controller to communicate with devices in the data plane.

XML is a markup language that is similar to HTML in structure; it uses tags to define blocks of data. Whereas HTML is used to render information on a webpage, XML is a more structured language that is used to format data in a way that can be easily transmitted over the Internet and parsed by a variety of applications.

JSON is a REST API encoding format. However, JSON returns data in the form of an object that contains key and value pairs. A single JSON object can contain multiple key and value pairs. Each key and value pair inside a JSON object is separated from the others by a comma (,). Furthermore, each pair's key is separated from its value by a colon (:). The element in quotation marks on the left side of each colon is the key. The element on the right side of each colon is the value, which might or might not be enclosed in quotation marks. There are several data value types that can be returned in JSON output: text, numeric, array, object, Boolean, and null.

YAML Ain't Markup Language (YAML) is not a REST API encoding format. YAML is a data serialization language that presents information in a format that is typically more human-readable than either XML or JSON. YAML is commonly used by the Ansible configuration management tool to store configuration playbooks.

Binary JSON (BSON) is not a REST API encoding format. BSON is a data serialization format that stores JSON data in a binary form that is not human-readable. This is in contrast to the text format that is typical of JSON. BSON is typically used in information storage systems, such as MongoDB.

Day 61 - REST APIs

Included Exam D items: Q14

Original Exam D Question 14

Which of the following statements is true about the HTTP GET method? (Select the best answer)

- A. It can be used to implement a webhook.
- B. It can be used to apply partial modifications to an object.
- C. It can be used to remove an object.
- D. It can be used to retrieve data from an object.

Correct Answer: D

Explanation

The Hypertext Transfer Protocol (HTTP) GET method can be used to retrieve data from an object. The following HTTP methods are most commonly used in Application Programming Interfaces (APIs):

- © GET - retrieves data from an object
- © PUT - adds information to an object
- © POST - creates an object
- © DELETE - removes an object
- © PATCH - applies partial modifications to an object

Although the HTTP POST method is most often used to create a new object or resource, it can also be used to implement a webhook. A webhook, also known as a reverse API, is a user-defined HTTP callback. Webhooks are event-driven rather than request-driven: instead of an application continuously requesting data, the application can wait for a particular event to occur before it reacts. Webhook notifications can be used on many Cisco platforms, including Meraki and Webex Teams.

Day 62 - Software-Defined Networking

Included Exam D items: Q6, Q10, Q37

Original Exam D Question 6

Which of the following Cisco DNA Center components is most likely to use REST? (Select the best answer.)

- A. the overlay
- B. the southbound API
- C. the underlay
- D. the northbound API

Correct Answer: D

Explanation

The northbound Application Programming Interface (API) is the Cisco Digital Network Architecture (DNA) Center component that is most likely to use Representational State Transfer (REST). Cisco DNA Center, which has been rebranded as Cisco Catalyst Center, abstracts and simplifies the complexity of network configuration by implementing a centralized controller and graphical user interface (GUI) to automate the configuration of network devices.

Northbound APIs enable the Cisco DNA Center controller to communicate with applications in the application plane. Applications use northbound APIs to send requests or instructions to a Software-Defined Networking (SDN) controller, such as the Cisco DNA Center controller, which uses that information to modify and manage network flow. REST is a northbound API architecture that uses Hypertext Transfer Protocol (HTTP) or HTTP Secure (HTTPS) to enable external resources to access and make use of programmatic methods that are exposed by the API. REST APIs typically return data in either Extensible Markup Language (XML) or JavaScript Object Notation (JSON) format.

The southbound API is not likely to use REST. A Cisco DNA controller uses two different sets of APIs: one set to communicate with applications and another set to communicate with devices in the data plane. A southbound API, such as NETCONF, can be used to enable the configuration of network devices in the data plane by the controller. NETCONF uses XML and Remote Procedure Calls (RPCs) to configure network devices. XML is used for both data encoding and protocol messages. NETCONF typically relies on Secure Shell (SSH) for transport.

The overlay is not likely to use REST. The overlay network is the Cisco Software-Defined Access (SDA) component that creates Virtual Extensible local area network (VXLAN) tunnels between Cisco SDA switches. The tunnels send and receive traffic between fabric endpoints. End-user subnets and endpoints are part of the overlay network.

The underlay is not likely to use REST. The underlay network is a more traditional network configuration of switches. It is a collection of devices, interfaces, and media that comprises the physical network that connects each fabric node. The underlay network is part of a dynamic discovery process that is involved in creating the overlay network's VXLAN tunnels. When an endpoint in a Cisco SDA network sends traffic to another endpoint, the traffic flows from the endpoint through the overlay network's VXLAN tunnels. The fabric is the entirety of the overlay network and the underlay network in a Cisco SDA network.

Original Exam D Question 10

Drag each network management technique to its description. Use all techniques. Each technique can be used only once.

Techniques Descriptions

Correct Answer: Output of show commands -> is often interpreted by a human administrator
Network automation -> uses a centralized controller to define data models
Automation script -> is written by a human to parse show command output

Explanation

Network automation uses a centralized controller to define and formalize data models. In addition, network automation aids reliable and consistent deployment of device configurations and software updates throughout an enterprise. Network automation includes the Software-Defined Networking (SDN) architecture. An SDN architecture is one in which management software is used to centralize device intelligence.

SDNs use northbound Application Programming Interfaces (APIs) to send network instructions from software applications to the central controller. Southbound protocols, which connect to a network's physical devices, are typically linked to the SDN by using a service abstraction layer (SAL). The SAL is a database, or registry, of the services provided by the southbound APIs. The APIs are bound to the registry so that the SAL can service an application's request.

The output of show commands is often interpreted by a human administrator. The issuing of Cisco IOS show commands to verify configurations and to troubleshoot networks is an action traditionally performed by an administrator. The administrator must then interpret the output of a show command to verify a configuration or to uncover problems with the configuration. Often, the output of a given show command must be compared to the output of other show commands or to a configuration standard in order for the administrator to obtain complete information.

An automation script is written by a human to parse show command output. Automation scripts are typically human-created programs that parse the output of show commands in order to obtain specific information about a specific configuration. Although they might ease administrative burden for common and repetitive tasks, automation scripts are less robust than network automation.

Original Exam D Question 37

In a controller-based network, which of the following typically contains end devices and Layer 2 switches? (Select the best answer.)

- A. the data plane
- B. the control plane
- C. the application plane
- D. the management plane

Correct Answer: A

Explanation

In a controller-based network, such as a Software-Defined Networking (SDN) network, the data plane typically contains end devices and Layer 2 switches. Layer 3 switches also often operate in the data plane. A centralized controller connects to the data plane by using a southbound Application Programming Interface (API), such as NETCONF, OpenFlow, OpFlex, or OnePK. Network tasks that are typically performed in the data plane include the encapsulation and decapsulation of packets, the adding or removing of trunk headers, the matching of Media Access Control (MAC) addresses to a MAC address table, the matching of Internet Protocol (IP) addresses to paths in a routing table, the encryption of data, Network Address Translation (NAT), and filtering by using either access control lists (ACLs) or port security.

In a controller-based network, the control plane is centralized. The control plane is responsible for network decision making in both a controller-based network and a traditional network. However, the control plane in a traditional network is typically distributed among many devices. The Open Shortest Path First (OSPF) routing protocol running on a series of routers on a traditional network is one example of a traditional control plane. OSPF makes routing decisions for packets that require routing among Layer 3 devices. In a controller-based network, the decision-making logic for packet-handling policies is either moved to a central controller or monitored by a central controller.

In a controller-based network, a centralized controller connects to the application plane by using a northbound API. The application plane is the component of a controller-based network in which applications that are written to allow interaction with the centralized controller reside. These applications are typically designed to improve network management efficiency through network

automation. A controller communicates with applications in the application plane by using a northbound API such as Representational State Transfer (REST) or Java Open Services Gateway initiative (OSGi).

In both a controller-based network and a traditional network, the management plane consists of network management protocols, such as Telnet, Secure Shell (SSH), Simple Network Management Protocol (SNMP), and Syslog. All of these protocols enable an administrator to connect to and manage a network device.

Day 63 - Ansible, Puppet, Chef & Terraform

Included Exam D items: Q62

Original Exam D Question 62

What type of file does Ansible use to specify the actions that will be performed on a client device? (Select the best answer)

A. aresource file

O 8. aconfiguration file

C. atemplate

D. amodule

E. aninventory

F. aplaybook

Correct Answer: F

Explanation

Ansible uses a playbook to specify the actions that will be performed on a client device. Ansible is a configuration management tool that operates on macOS, on Linux, or within a Linux virtual machine (VM) on Windows. Unlike other configuration management software packages, such as Terraform, Ansible does not use agent software on managed nodes. Instead, Ansible uses Secure Shell (SSH) to communicate with clients. Clients that are managed by Ansible are agentless, which means that no code needs to be preinstalled on the client device. Playbooks are written in YAML Ain't Markup Language (YAML).

An Ansible inventory is a file that contains a list of client device host names and other device information. By default, Ansible will push configurations to client devices. However, you can configure this capability by installing a program called Ansible-pull in order to allow clients to pull configurations from Ansible.

An Ansible template is a file that is written in Jinja2 and contains the configuration that will be used on the client. However, because each client is configured differently, the template contains defined variables. For example, the following template might be used to configure a host device:

```
hostname {{host}}
```

```
f
```

```
interface fastethernet1/1
```

```
ip address {{ipaddr1}} {{submask1}}
!
```

```
interface fastethernet1/2
```

```
ip address {{ipaddr2}} {{submask2}}
```

Ansible uses a variables file that will substitute values into the template variable definitions. The following is a sample Ansible variable file that is written in YAML:

```
host: RouterA
```

```
ipaddr1: 192.168.1.1  
submask1: 255.255.255.0  
ipaddr2: 192.168.2.1  
submask2: 255.255.255.0
```

Ansible does not use configuration files, resource files, or modules; these files are used by Terraform. Terraform is an infrastructure as code (IaC) configuration management tool that can be used to manage cloud-based and local devices. A Terraform configuration file is used to configure the devices. A Terraform resource file specifies the devices, networks, or data stores that you want to manage. Terraform modules are used to organize these configuration and resource files.